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Non-negative matrix factorization and deconvolution as a dual simplex problem

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Abstract

Background: Non-negative matrix factorization is a powerful linear algebra tool used in multiple areas of data analysis, including computational biology. Despite numerous optimization methods devised for non-negative matrix factorization, our understanding of the inherent topological structure within factorizable matrices remains limited.

Results: This study reveals the topological properties of linear mixture data, leading to a remarkable reduction of the non-negative matrix factorization optimization problem to a search for $K(K-1)$ variables, where K represents the number of pure components, regardless of the initial matrix size. This is achieved by revealing complementary simplex structures existing in both feature and sample spaces and leveraging the Sinkhorn transformation to find the relationship between these simplexes. We validate this approach in the context of an unconstrained mixed images scenario and achieve a significant improvement in decomposition accuracy. Furthermore, we successfully applied the proposed approach in the biological context of bulk RNA-seq gene expression deconvolution.

Conclusions: The Dual Simplex unified analytical framework improves robustness to noise and enhances optimization stability, enabling accurate recovery of component proportions and expression profiles. Importantly, the framework naturally accommodates both reference-free and marker-based deconvolution settings, providing a general and efficient solution for analyzing complex biological mixtures such as bulk RNA-seq and single-cell derived data.

Keywords: NMF, Simplex, Deconvolution, Unmixing, Clustering, ScRNA-seq, Gene expression

Background

Nonnegative matrix factorization (NMF) [1] is widespread in a variety of computational biology problems, including genomics [2], transcriptomics [3, 4], mass spectrometry [5], and cytometry [6] etc. It is also well-known beyond biological sciences as a common technique for feature extraction [7] and data compression [8]. NMF seeks to represent a non-negative $M \times N$ matrix V as a product of two non-negative matrices W and H ($V = W \times H$). While the data matrix V is typically high-dimensional (N ,



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$M \sim 100\text{--}1000\text{s}$), its rank (K) is often much smaller in biological settings ($\sim 10\text{s}$). The rank (K) of matrices W and H is usually thought of as the number of independent components/clusters/features comprising the dataset [9]. In these terms, W is the *feature matrix* representing the independent components of the data (i.e., main variational components encoding the data columns), and H is the *coefficients matrix* representing the contributions of the independent components for each mixture/sample (i.e., columns of the matrix V).

One of the most common applications of NMF is the linear unmixing/deconvolution problem, where matrix H is a proportion matrix, i.e., sum-to-one normalized [10]. In this setting, W represents a set of K pure samples (i.e., components or original sources), and each column of the mixture matrix V is modeled as a linear combination of these sources with proportions given by H [11]. This constrained problem formulation is also widespread in modern computational biology – spanning the analysis of spectral cytometry data [12], spatial expression data analysis [13], and the dissection of the composition of bulk transcriptional profiles [14]. Over the last decade, multiple computational approaches have been implemented to solve the deconvolution problem, especially in the context of gene expression, ranging from support vector machines [15, 16] to deep neural networks [17, 18]. While these methods often employ sophisticated computational techniques, no unified understanding of the underlying data structure for factorizable matrices exists at the moment. Indeed, even though NMF or deconvolution problems are seeking low-dimensional representations relative to the dimensionality of the input data matrix, there is no explicit mathematical procedure that could rigorously leverage this notion in the factorization procedure.

In this work, we introduce an analytical framework that reveals dual/complementary simplexes within the feature and sample spaces of factorizable matrices and establishes their direct relationship. This connection and subsequent analysis of these simplexes allow the NMF problem of finding W and H to be formulated as identifying simplex vertices in low-dimensional spaces, yielding both conceptual clarity and computational efficiency.

We begin by examining the topological properties of a nonnegatively factorizable matrix. First, we note that a non-negative $M \times N$ matrix V can be represented as N samples in an M -dimensional “feature” space or as M features within an N -dimensional “sample” space. Normalizing the columns of V embeds the N samples within a $(K - 1)$ -dimensional simplex in the feature space, where K is the rank of V with $K \ll M$ and N . Similarly, normalizing the rows embeds the M features within a $(K - 1)$ -dimensional simplex in the sample space.

Next, we describe how a sequence of alternating row and column normalizations (i.e., the Sinkhorn transformation) of V leads to two representations of it: $\overset{\infty}{V}_{fs}$ in the feature space and $\overset{\infty}{V}_{ss}$ in the sample space. These two forms are analytically and simply related by the ratio of dimensions, M/N . Using this relationship, we prove that the SVD of either the $\overset{\infty}{V}_{fs}$ or $\overset{\infty}{V}_{ss}$ yields singular vectors that optimally define the projection of these matrices onto K -dimensional hyperplanes of their respective simplexes. This is the main theoretical result of this work.

We further demonstrate that the simplex vertices within these low-dimensional spaces correspond to projections of two non-negative matrices: $\overset{\infty}{H}_{ss}$ for the sample space and $\overset{\infty}{W}_{fs}$ for the feature space. These projections define the factorizations $\overset{\infty}{V}_{ss} = \overset{\infty}{W}_{ss} \times \overset{\infty}{H}_{ss}$ and $\overset{\infty}{V}_{fs} = \overset{\infty}{W}_{fs} \times \overset{\infty}{H}_{fs}$, which can be used to reconstruct matrices W and H , that optimally approximate the nonnegative factorization $V = W \times H$.

We incorporated these theoretical results in a four-step approach, which we call Dual Simplex. For any non-negative matrix V , this approach involves: 1) Performing the Sinkhorn transformation of the matrix V ; 2) Projection of both the rows and columns of the Sinkhorn transformed matrix into the respective SVD subspaces; 3) Simultaneously identifying the simplex vertices in the projected spaces, 4) Applying the reverse Sinkhorn transformation.

Finally, we demonstrate the power of the Dual Simplex approach through applications to image unmixing, deconvolution of bulk transcriptional profiles, and clustering of single-cell RNA-seq data. These results advance theoretical understanding of the NMF problem by providing unique insights into the topological structures underlying non-negative factorizable matrices. In practice, the resulting reduction in dimensionality greatly simplifies the computational task and provides guidance for the design of unsupervised (complete) and supervised (partial or marker-based) deconvolution methods, as well as for broader NMF applications such as image decomposition and spatial data analysis.

Results

Complementary feature and sample simplex structures

Let's consider the non-negative matrix factorization problem where $V = W \times H$. In general terms, a non-negative factorizable matrix V can be viewed as a collection of N (e.g., number of samples) dots in an M -dimensional space (e.g., number of features). We will call this space a *feature space* since individual axes correspond to the values of the features (Fig. 1a). In turn, V^T can be viewed as a collection of M dots (each representing individual features) in an N -dimensional space. We will call this space a *sample space* (Fig. 1a). In the most general case, when matrix V is neither row- nor column-normalized, matrices V and V^T represent unorganized collections of points in their corresponding spaces (Fig. 1a).

It has been observed that a row normalization procedure applied to any positive and factorizable matrix leads to the emergence of a simplex structure in the space of its columns [19]. Specifically, the row-normalized and transposed matrix $\tilde{V}^T$ places all feature-dots within the $(K - 1)$ -dimensional simplex inside the N -dimensional space (Fig. 1b, Additional file 1: Supplementary Note 1). In fact, for factorization $\tilde{V} = \tilde{W} \times \tilde{H}$, the geometrical vertices of the simplex correspond to the rows of the matrix $\tilde{H}$ within the space of vectors $\tilde{W}^T$ (Fig. 1b). Moreover, the closer a feature point lies to a simplex vertex, the more specific it is that component – that is, the feature has high values in that component but relatively low values in others (this is the same as Well-Grounded Points [20])

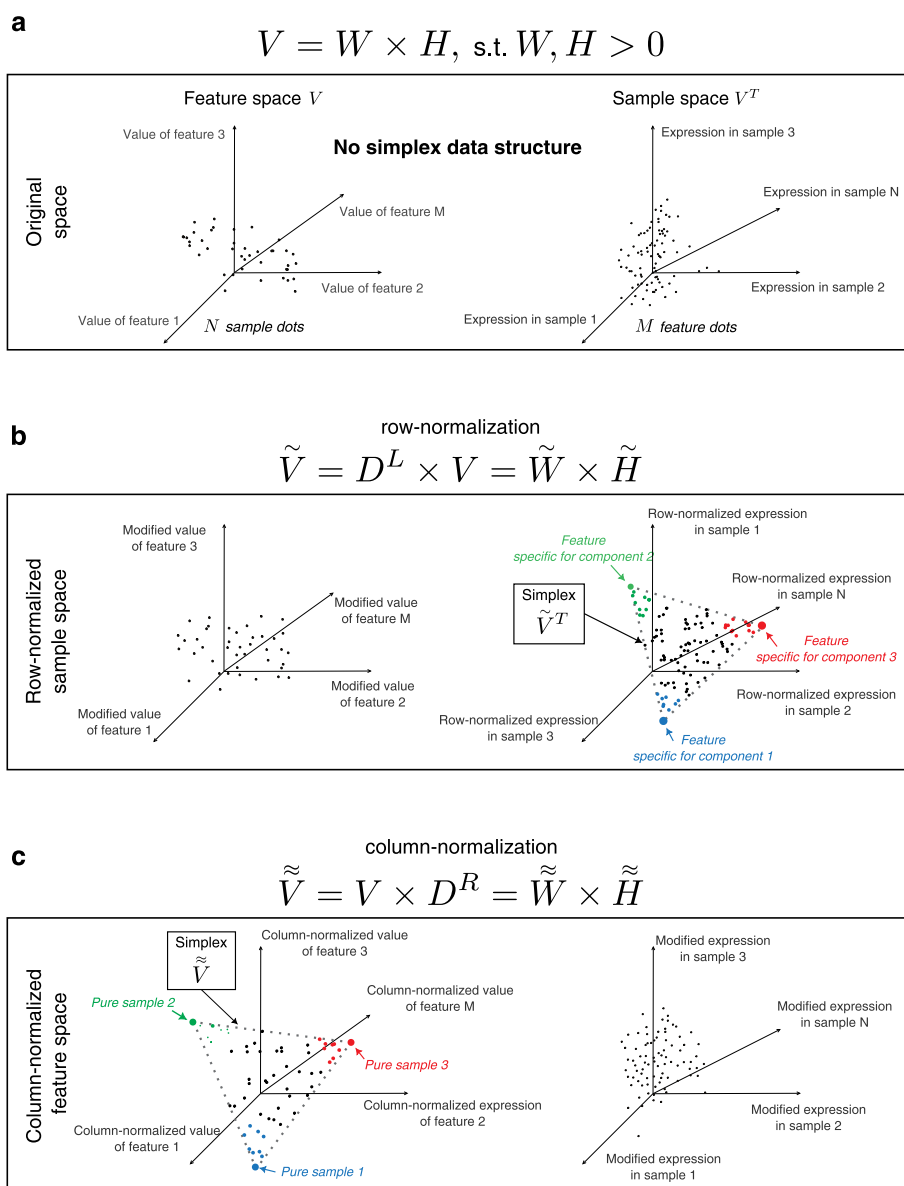


Fig. 1 Existence of complementary features and samples simplex structures for normalized matrices. **a** Schematic representation of the data matrix in feature and sample spaces. **b** Row normalization and transposition of the original matrix aligns feature dots within a simplex in a sample space. **c** Column normalization aligns sample dots within a simplex in feature space

(Fig. 1b). Accordingly, finding simplex vertices imputed from the position of component-specific features in the sample space is equivalent to solving the deconvolution problem via finding the matrix $\tilde{H}$ (e.g., [19]). We and others demonstrated the application of this approach to the gene expression data deconvolution problem [21, 22]. For example, in the case of a mixture of 3 types, this results in a 2-dimensional simplex (i.e., triangle) embedded in the multidimensional space of all the samples [21].

On the other hand, column normalization of any positive and factorizable matrix reveals the simplex structure in the complementary feature space [23]. Specifically, column normalization of the factorizable matrix V results in the matrix $\tilde{V}$ that describes a $(K - 1)$ -dimensional simplex in an M -dimensional space (Fig. 1c, Additional file 1: Supplementary Note 2 for proof). Indeed, $\tilde{V}$ can be viewed as a collection of N dots (each representing an individual sample) in the M -dimensional space of features (Fig. 1c). In this space, the positions of the sample dots have changed, acquiring a new geometrical structure (Fig. 1a, c). In the context of gene expression deconvolution, the vertices of this simplex correspond to the transcriptional profiles of pure samples, i.e., columns of the matrix $\tilde{W}$ (Additional file 2: Supplementary Note 2). The closer a sample point lies to a vertex, the purer its composition. Thus, identifying simplex vertices in this space is also equivalent to solving the deconvolution problem via finding matrix $\tilde{W}$.

Broadly speaking, the existence of these two simplex structures reflects two alternative approaches to solving the NMF problem: either by finding simplex vertices in the feature space or in the sample space (Additional file 2: Fig. S1). The choice of space is, in principle, arbitrary and may be guided by how well the simplex structure is represented, which depends on the presence of sufficient pure or high-proportion samples in the sample space, or sufficiently component-specific features in the feature space [24]. Moreover, as we demonstrate below, there exists a connection between these simplexes that provides us with a unique and powerful approach to solving the NMF/deconvolution problem within a subspace of significantly lower dimensionality.

The Sinkhorn transformation of the data matrix links features and samples simplexes

First, let us remark that since column normalization leads to a simplex for any factorizable matrix (Fig. 1), the transposition and column normalization of the previously transformed matrix $\tilde{V}^T$ also results in a matrix, which forms a simplex in a *feature space*. We can formalize this transformation using sequential multiplication by diagonal normalization matrices alternating between left and right multiplication (Eq. 1):

$$\tilde{\tilde{V}} = \tilde{V} \times D_{v_1} = D_{v_0} \times V \times D_{v_1} \quad (1)$$

where D_{v_0}, D_{v_1} are diagonal matrices: D_{v_0} is a row-normalizing matrix for V and D_{v_1} is a column normalizing matrix for the row-normalized matrix $\tilde{V}$.

Next, we note that subsequent row normalization, applied to the matrix $\tilde{\tilde{V}}$ (Eq. 1), will result in a matrix $\tilde{\tilde{\tilde{V}}}$ representing a simplex in a *sample space* which will be different from the previous simplex in the *sample space* obtained in the first step, i.e. $\tilde{\tilde{\tilde{V}}} = D_{v_3} \times \tilde{\tilde{V}} \neq \tilde{V}$. The next iteration will yield another simplex structure in the *feature space*, etc. In essence, the consecutive application of row normalization followed by column normalization conserves the simplex structures in their respective spaces. Such an iterative normalization process is also known as *the Sinkhorn transformation* of a matrix [25, 26] (Fig. 2a). Notably, according to the Sinkhorn-Knopp theorem [25], the odd and even elements of this sequence will converge towards two related matrices $\tilde{\tilde{\tilde{\tilde{V}}}}^\infty$ (the row-normalized matrix of the *sample space* simplex) and $\tilde{\tilde{\tilde{\tilde{V}}}}^\infty$ (the column-normalized matrix of the *feature space* simplex):

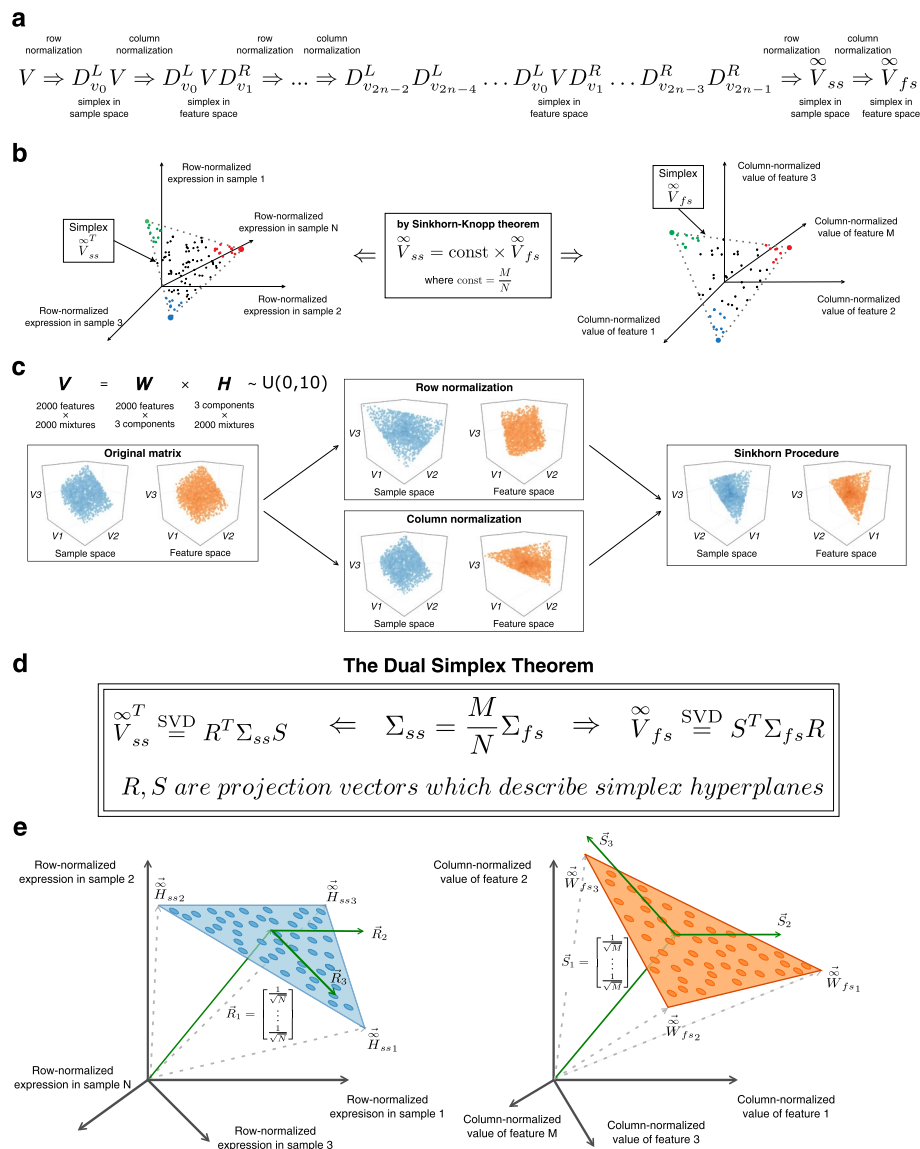


Fig. 2 The Sinkhorn transformation enables the identification of the simplex hyperplane projection vectors. **a** Sinkhorn transformation is a process of iterative left and right multiplication by diagonal matrices (i.e., row and column normalization), producing two converging sequences of matrices, representing simplexes in feature and sample spaces. **b** Converged matrices in feature and sample spaces are proportionally related. **c** Example of Sinkhorn transformation applied to randomly generated factorizable matrix ($K=3$, $M=2000$, $N=2000$). Matrices W and H sampled from a univariate uniform distribution with values ranging from 0 to 10. While original spaces of features and samples do not contain simplex structures (left), column and row normalizations reveal a simplex structure for one of the spaces (middle). Iterative normalizations applied to the matrix align points within simplexes for both spaces (right). **d** The Dual Simplex theorem formulation. Singular vectors of Sinkhorn-transformed matrices provide projection vectors to hyperplanes in which samples and feature simplexes are located. **e** Corollary of the Dual Simplex theorem: due to normalization constraint R_1 and S_1 have fixed form and represent the shift of the entire simplex hyperplane from the zero point

$$\begin{array}{c}
 \begin{array}{c}
 \text{row} \\ \text{norm}
 \end{array} \xrightarrow{\quad} V \xrightarrow{\quad} \underbrace{D_{v_0} \times V}_{\substack{(1) \\ \sim V \text{ or } V_{ss} \\ \text{simplex in} \\ \text{sample space}}} \xrightarrow{\text{column norm}} \underbrace{D_{v_0} \times V \times D_{v_1}}_{\substack{(2) \\ \approx V \text{ or } V_{fs} \\ \text{simplex in} \\ \text{feature space}}} \xrightarrow{\text{row norm}} \dots \xrightarrow{\text{column norm}} \\
 \xrightarrow{\text{column norm}} \underbrace{D_{v_{2n-2}} \times D_{v_{2n-4}} \dots \times D_{v_0} \times V \times D_{v_1} \times \dots \times D_{v_{2n-3}} \times D_{v_{2n-1}}}_{\substack{(2n) \\ V_{fs} \\ \text{simplex in} \\ \text{feature space}}} \xrightarrow{\text{row norm}} \overset{\infty}{V}_{ss} \xrightarrow{\text{column norm}} \overset{\infty}{V}_{fs}.
 \end{array}$$

Furthermore, elements of these matrices will be proportional to each other [25]. In the case of an initial $M \times N$ matrix V , the proportionality constant is M/N (Fig. 2b, Additional file 1: Supplementary Note 3):

$$\overset{\infty}{V}_{ss} = \frac{M}{N} \overset{\infty}{V}_{fs} \quad (2)$$

This convergence can be illustrated in a simulation scenario, where normalization is applied to the product of two randomly generated matrices W and H ($M = 2000, N = 2000, K = 3$) (Fig. 2c left). While single row or column normalization results in a single simplex structure within sample and feature space, respectively, the Sinkhorn procedure of as few as 3 iterations, reveals the simplex structures within both spaces simultaneously (Fig. 2c right).

The Sinkhorn transformation enables simultaneous identification of dual simplex hyperplanes

We will now consider how corresponding simplices can be identified within $\overset{\infty}{V}_{ss}^T$ and $\overset{\infty}{V}_{fs}$. Locating the data simplex generally involves two fundamental steps: first, finding a lower-dimensional hyperplane wherein the simplex is located, and then proceeding to subsequent identification of the simplex vertices within this hyperplane (Additional file 2: Fig. S1). We find that the Sinkhorn transformation enables a very elegant identification of the hyperplanes in which simplexes are confined.

To demonstrate this, we will first consider independent problems of hyperplane identification for features and samples simplexes and then describe the relationship between them. Broadly, considering that the rank of the matrix V is K , we approach the task of hyperplane identification as a problem of searching for a projection onto the corresponding K -dimensional subspace for either row-normalized and transposed matrix $\tilde{V}^T$ or the column-normalized matrix $\tilde{V}$. One can utilize a projective NMF framework searching for respective projection operators $Proj_R = R^T R$ and $Proj_S = S^T S$ [8]. Here, R is a $(K \times N)$ -matrix comprising orthogonal vectors that define the coordinate system within a projected subspace of the sample space, and S is a $(K \times M)$ -matrix, defining the coordinate system within a projected subspace of the feature space (Eq. 3, Additional file 2: Fig. S2a, Additional file 1: Supplementary Note 4, 5).

$$R' = \underset{R}{\operatorname{argmin}} \| \tilde{V}^T - R^T \times R \times \tilde{V}^T \| \quad S' = \underset{S}{\operatorname{argmin}} \| \tilde{V} - S^T \times S \times \tilde{V} \| \quad (3)$$

s. t. $R \times R^T = \mathbb{I}_{K \times K}$ s. t. $S \times S^T = \mathbb{I}_{K \times K}$

This K -dimensional optimization problem at hand is recognized to be equivalent to singular value decomposition and can be solved for any non-negative matrix V of rank K [27].

However, the formulation presented in (Eq. 3) does not directly identify the hyperspace containing the $(K - 1)$ -dimensional simplex, as singular vectors, in general, may not align precisely with this hyperspace (Additional file 2: Fig. S2a). Ideally, we aim to identify a projection operator that would act on $\tilde{V}^T$ or $\tilde{V}$ and yield a $(K - 1)$ -dimensional simplex. This decrease in dimensionality stems from an additional sum-to-one constraint, which is a consequence of the normalization procedure. In principle, one can identify such a $(K - 1)$ -dimensional structure embedded in a K -dimensional space by initially getting a centered data matrix $\tilde{V}_c^T = \tilde{V}^T - \frac{\sum_{j=1}^M \tilde{V}_{sj}^T}{M} \mathbb{I}_M$ ($\tilde{V}_c = \tilde{V} - \frac{\sum_{j=1}^N \tilde{V}_{sj}}{N} \mathbb{I}_N$), where $\mathbb{I}_M = (1, 1, \dots, 1) \in \mathbb{R}^M$, $\mathbb{I}_N = (1, 1, \dots, 1) \in \mathbb{R}^N$, and then by proceeding with the finding of the $(K - 1)$ -dimensional singular value decomposition (SVD) of the centered matrix [23] (Additional file 2: Fig. S2b).

$$R'' = \underset{R}{\operatorname{argmin}} \| \tilde{V}_c^T - R^T \times R \times \tilde{V}_c^T \| \quad S'' = \underset{S}{\operatorname{argmin}} \| \tilde{V}_c - S^T \times S \times \tilde{V}_c \| \quad (4)$$

s. t. $R \times R^T = \mathbb{I}_{(K-1) \times (K-1)}$ s. t. $S \times S^T = \mathbb{I}_{(K-1) \times (K-1)}$

Note that the solution vectors for (Eq. 4), taken together with centering vectors $\vec{R}_c(\vec{S}_c)$, form two sets of vectors $(\vec{R}_c, \vec{R}_2, \dots, \vec{R}_K)$ and $(\vec{S}_c, \vec{S}_2, \dots, \vec{S}_K)$ that can describe the hyperplane in the space of initial features and samples. However, these two sets of vectors are not necessarily mutually orthogonal. For instance, $\vec{R}_c$ is typically not orthogonal to the hyperspace formed by $(\vec{R}_2, \dots, \vec{R}_K)$ (Additional file 2: Fig. S2b). Therefore, operators formed by $(\vec{R}_c, \vec{R}_2, \dots, \vec{R}_K)$ and $(\vec{S}_c, \vec{S}_2, \dots, \vec{S}_K)$ do not form projection operators acting on $\tilde{V}^T$ and $\tilde{V}$ respectively.

Appropriate projection vectors can be obtained through an orthogonalization procedure applied to $\vec{R}_c(\vec{S}_c)$, or equivalently, by introducing the orthogonalization constraint into the optimization problem (Eq. 5, Additional file 1: Supplementary Note 6, 7):

$$R^* = \underset{R}{\operatorname{argmin}} \| \tilde{V}^T - R^T \times R \times \tilde{V}^T \| \quad S^* = \underset{S}{\operatorname{argmin}} \| \tilde{V} - S^T \times S \times \tilde{V} \| \quad (5)$$

s. t. $R \times R^T = \mathbb{I}_{K \times K}$ s. t. $S \times S^T = \mathbb{I}_{K \times K}$,

$\tilde{V} \vec{R}_1 = c \mathbb{I}_M$ $\tilde{V}^T \vec{S}_1 = c \mathbb{I}_N$

or in case of Sinkhorn transformed matrices (Eq. 6, Additional file 1: Supplementary Note 8):

$$R^* = \underset{R}{\operatorname{argmin}} \| \tilde{V}_{ss}^T - R^T \times R \times \tilde{V}_{ss}^T \| \quad S^* = \underset{S}{\operatorname{argmin}} \| \tilde{V}_{fs} - S^T \times S \times \tilde{V}_{fs} \| \quad (6)$$

s. t. $R \times R^T = \mathbb{I}_{K \times K}$ s. t. $S \times S^T = \mathbb{I}_{K \times K}$,

$\tilde{V}_{ss} \vec{R}_1 = c \mathbb{I}_M$ $\tilde{V}_{fs}^T \vec{S}_1 = c \mathbb{I}_N$

In the context of this optimization problem (Eq. 6), the special properties of Sinkhorn transformed matrices enable a very elegant identification of the hyperplanes in which

simplexes are confined. Specifically, the singular value decomposition of the matrix $\overset{\infty}{V}_{ss}$ (or $\overset{\infty}{V}_{fs}$) yields the sought-after projection vectors R and S for the constrained projection optimization problem (Fig. 2d, Additional file 1: Supplementary Note 9 for proof). This represents the key theoretical result of our work, which is formulated as the Dual Simplex Theorem:

Theorem. For Sinkhorn transformed matrices $\overset{\infty}{V}_{ss}$ $\overset{\infty}{V}_{fs}$ (where $\overset{\infty}{V}_{ss} = \frac{M}{N} \overset{\infty}{V}_{fs}$) obtained from the non-negative matrix V , solution vectors to optimization problems (6) can be found as follows:

$$\overset{\infty}{V}_{ss}^T \stackrel{\text{SVD}}{=} R^T \Sigma_{ss} S \text{ or } \overset{\infty}{V}_{fs} \stackrel{\text{SVD}}{=} S^T \Sigma_{fs} R$$

- R^T could be found as the left singular matrix of $\overset{\infty}{V}_{ss}^T$ and the right singular vectors of $\overset{\infty}{V}_{fs}$
- S^T could be found as the right singular matrix of $\overset{\infty}{V}_{ss}$ and the left singular vectors of $\overset{\infty}{V}_{fs}$

The simple consequence of this Theorem and (Eq. 2) is that singular values are linearly connected for matrices $\overset{\infty}{V}_{ss}$ and $\overset{\infty}{V}_{fs}$, so that $\Sigma_{ss} = \frac{M}{N} \Sigma_{fs}$. In short, the Sinkhorn transformation and subsequent SVD representation offer a clear path to identifying each of the individual hyperplanes where simplexes are located.

Geometrical implications of the Dual Simplex Theorem

Furthermore, from the Dual Simplex Theorem and the matrix row/column normalization property, it follows that the pair of the top singular vectors $\vec{R}_1^T$ and $\vec{S}_1^T$ has well-defined forms and singular values. Specifically, vector $\vec{R}_1^T$ is $\left[\frac{1}{\sqrt{N}} \dots \frac{1}{\sqrt{N}} \right]$ and corresponds to the singular value $\sigma_{ss,1} = \sqrt{M/N}$ (Fig. 2e, Additional file 1: Supplementary Note 9 for proof). We demonstrate that the $(K - 1)$ -dimensional simplex of sample space lies in the hyperplane that is strictly perpendicular to the vector $\vec{R}_1^T$, i.e. diagonal in the N -dimensional space, and intersects this diagonal at a specific point ($\vec{R}_1^T$). Similarly, the corresponding $(K - 1)$ -dimensional hyperplane of feature space simplex is strictly perpendicular to the M -dimensional diagonal $\vec{S}_1^T = \left[\frac{1}{\sqrt{M}} \dots \frac{1}{\sqrt{M}} \right]$, which is a singular vector corresponding to the singular value $\sigma_{fs,1} = \sqrt{N/M}$. These vectors ($\vec{S}_1^T$ and $\vec{R}_1^T$) are present in the singular value decomposition for any Sinkhorn-transformed data matrix and are a consequence of sum-to-one scaling iterations. Geometrically, these vectors represent the shift of the simplex hyperplane away from the center of coordinates along the diagonal (Additional file 1: Supplementary Note 9 for proof). The remaining $(K - 1)$ vectors of R (and S) define the subspace of the $(K - 1)$ -dimensional simplex itself. For instance, in the case of 3 cell types ($K = 3$), a 2-dimensional simplex (a triangle) is located within a subspace defined by vectors $\vec{R}_2^T, \vec{R}_3^T$ in the samples space and vectors $\vec{S}_2^T, \vec{S}_3^T$ in the features space (Fig. 2e). Understanding the structure of R and S also allows interpretable low-dimensional representations of the Sinkhorn transformed coefficient (H) and pure components (W) matrices (matrices $\overset{\infty}{H}_{ss}$ and $\overset{\infty}{W}_{fs}$) as corresponding simplexes. This can be done by introducing $K \times K$ simplex matrices

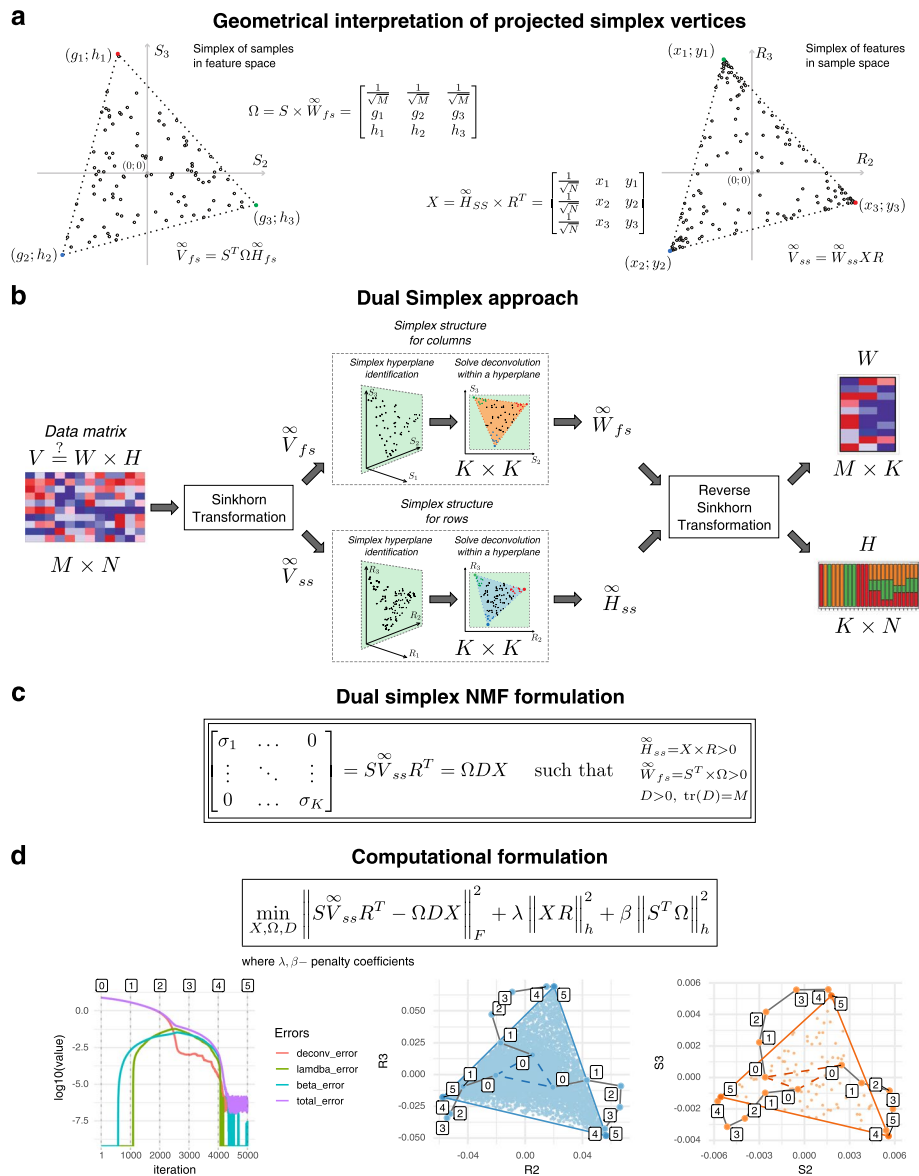


Fig. 3 Interpretation of the simplex coordinates. Unified Dual Simplex problem formulation. **a** The geometrical interpretation of the projected data points: due to the proportional relation between simplexes and constraints enforced for vectors R and S , the first coordinate of each projected point represents only a shift of the hyperplane away from the center of coordinates, while the entire simplex structure is described by only $(K-1)$ remaining coordinates. The search for simplex vertices is therefore a search for coordinates for a geometrical simplex in a $(K-1)$ -dimensional space (e.g., a triangle in the case when $K=3$). **b** Schematic illustration of the Dual Simplex approach. The input matrix V to be factorized into W and H is first Sinkhorn-transformed, producing two representations of the original data (V_{fs}^{∞} and V_{ss}^{∞}). Then, columns of V_{fs}^{∞} and rows of V_{ss}^{∞} are projected into the respective spaces of singular vectors. Within these spaces, an optimization is performed to identify the vertices of the simplexes. These vertices are then projected back into W_{fs}^{∞} and H_{ss}^{∞} , which then could be traced back to the original W and H using the reverse Sinkhorn procedure. **c** The Dual Simplex problem formulation is a K -dimensional matrix equation equivalent to the initial NMF problem. The left part of this equation is a precomputed singular values matrix, while the right part contains three unknown variables representing coordinates of simplexes in both feature and sample spaces, as well as the diagonal normalization matrix D , which connects these spaces. **d** The gradient descent optimization formulation of the Dual Simplex problem. The cost function comprises the main deconvolution term as well as two additional terms (with coefficients λ and β) to address the positivity constraints. The optimization trajectory and endpoints in feature and sample spaces for simulated gene expression data ($N = 10000$, $M = 100$, $K = 3$, σ_{noise}) are presented at the bottom

$X = \overset{\infty}{H}_{ss} \times R^T$ and $\Omega = S \times \overset{\infty}{W}_{fs}$, where $\overset{\infty}{V}_{ss} = \overset{\infty}{W}_{ss} \times \overset{\infty}{H}_{ss}$ and $\overset{\infty}{V}_{fs} = \overset{\infty}{W}_{fs} \times \overset{\infty}{H}_{fs}$ (Fig. 3a). Generally speaking, the rows of X and the columns of Ω define K points in K -dimensional space, which are located at the simplex vertices (Fig. 3a). In fact, as a consequence of the sum-to-one scaling operations and the corresponding form of the $\overset{\infty}{R}_1^T$ and $\overset{\infty}{S}_1^T$ vectors, matrices X and Ω will also be partially known. For example, for 3 cell types, out of 9 elements (more generally, $K \times K$), 3 elements of the matrix $X(\Omega)$ will be fixed (more generally, K elements) (Fig. 3a, Additional file 1: Supplementary Note 9 for general case proof). The first column of matrix X will be a column vector of $1/\sqrt{N}$, and the first row of matrix Ω will be a row vector of $1/\sqrt{M}$ (Fig. 3a, Additional file 1: Supplementary Note 9). Since rows of matrix X represent coordinates of the features' simplex in K -dimensional space of samples, the remaining $K - 1$ components of each row-vector of X will define simplex location in the $(K - 1)$ -dimensional space (Fig. 3a). Similarly, column-vectors of Ω define $(K - 1)$ -dimensional coordinates of samples' simplex in the feature space (Fig. 3a).

The Dual Simplex approach

With the exact projective spaces and topological structure of the factorizable matrix in hand, we can formulate the NMF problem as a four-step projection approach (Fig. 3b):

1) Sinkhorn transformation

Apply alternating row and column normalization to the original data matrix V to obtain scaled representations of the connected sample space and feature space simplexes (matrices $\overset{\infty}{V}_{ss}$ and $\overset{\infty}{V}_{fs}$).

2) Projection into SVD space

Project the transformed matrices into their respective SVD subspaces (R and S), yielding reduced-dimensional representations in which the encoded data points remain fully recoverable.

3) Identification of simplex vertices

Simultaneously identify the vertices of the simplexes (X and Ω), using the low-dimensional projected data. These vertices, upon reverse projection, define the factorization matrices $\overset{\infty}{H}_{ss}$ and $\overset{\infty}{W}_{fs}$, satisfying $\overset{\infty}{V}_{ss} = \overset{\infty}{W}_{ss} \times \overset{\infty}{H}_{ss}$ and $\overset{\infty}{V}_{fs} = \overset{\infty}{W}_{fs} \times \overset{\infty}{H}_{fs}$ (Additional file 1: Supplementary Note 10 for details).

4) Reverse Sinkhorn transformation

Using the relationship between the two simplexes, reconstruct the remaining factorization matrices $\overset{\infty}{W}_{ss}$ and $\overset{\infty}{H}_{fs}$ and track back all iterations of the scaling to the original solution matrices W and H , using normalization matrices obtained during the forward Sinkhorn procedure (Additional file 1: Supplementary Note 11 for details).

Unified Features and Samples Dual Simplex Optimization Problem

The search for individual simplex vertices – represented by the coordinate matrix X in the sample space or Ω in the feature space – can be formulated through two distinct optimization problems (Additional file 1: Supplementary Notes 4–7). However, since both simplexes coexist within the Sinkhorn-transformed data, it is natural to seek a unified treatment. Given that left and right singular vectors are not randomly

ordered but correspond to each other's variability components, the *Dual Simplex Theorem* indicates that there is a relationship between the hyperspaces of features and samples simplexes. In fact, the Sinkhorn transformation enables a unified optimization problem formulation for two simplexes simultaneously.

Since matrices $\tilde{V}_{ss}$ and $\tilde{V}_{fs}$ are proportional, we can merge the two projection formulations into a single one (Eq. 7):

$$\left. \begin{aligned} \tilde{V}_{ss} - \tilde{V}_{ss} \times R^T \times R &= 0 \\ \tilde{V}_{fs} - S^T \times S \times \tilde{V}_{fs} &= 0 \\ \tilde{V}_{fs} &= \tilde{V}_{ss} \times \frac{N}{M} \end{aligned} \right\} \Rightarrow \tilde{V}_{ss} = S^T \times S \times \tilde{V}_{ss} \times R^T \times R \quad (7)$$

This then could be represented in terms of factor matrices ($\tilde{V}_{ss} = \tilde{W}_{ss} \times \tilde{H}_{ss}$) and their projection coordinates ($\tilde{H}_{ss} \times R^T = X$, $\tilde{W}_{ss} = \tilde{W}_{fs} \times D$ and $S \times \tilde{W}_{fs} = \Omega$), where D is a diagonal non-negative matrix that connects matrices $\tilde{W}_{ss}$ and $\tilde{W}_{fs}$ with the property that $\text{tr}(D) = M$ (Additional file 1: Supplementary Note 8):

$$S \times \tilde{V}_{ss} \times R^T = S \times \tilde{W}_{ss} \times \tilde{H}_{ss} \times R^T \Rightarrow S \times \tilde{V}_{ss} \times R^T = \Omega \times D \times X \quad (8)$$

Equation 8 holds for all exactly factorizable non-negative matrices and can serve as the basis for optimization problems in noisy and approximation scenarios. In such a case, the main goal of the optimization is to minimize the difference between the left and right parts of this equation, thereby obtaining the approximate simplex coordinates (X, Ω) with the highest possible precision. As a result, we describe the following common optimization formulation for the simultaneous search of samples and features simplexes (Eq. 9, Fig. 3c, Additional file 1: Supplementary Note 8):

$$\begin{aligned} X, \Omega, D = \underset{\substack{X \\ \Omega \\ D}}{\text{argmin}} & \| S \times \tilde{V}_{ss} \times R^T - \Omega \times D \times X \| \\ \text{s. t. } & S^T \times \Omega > 0 \\ & X \times R > 0 \\ & D > 0 \end{aligned} \quad (9)$$

Note, that $\Sigma_{ss} = S \times \tilde{V}_{ss} \times R^T$ is a $K \times K$ diagonal matrix of singular values, which is precomputed based on the Sinkhorn transformed matrix (see Dual Simplex Theorem). Therefore, the NMF optimization problem (Eq. 9) becomes a problem of simultaneous optimization of two simplexes (Ω and X) in a low $(K - 1)$ -dimensional subspaces, along with a diagonal matrix D , which represents linear scaling between two simplexes (Fig. 3c).

This problem can be tackled using a gradient descent numerical approach (Fig. 3d, [Methods](#), Additional file 1: Supplementary Note 10). Using gene expression simulated data with added noise as a simple illustrative study case ([21], [Methods](#)), we find our computation procedure stably converging, producing valid results for the location of simplex vertices (Fig. 3d).

Taken together, the proposed framework (Figs. 1, 2, 3) describes a projective approach to the general non-negative matrix factorization formulation and solution, which we called Dual Simplex. This approach could be applied to any non-negative matrix to

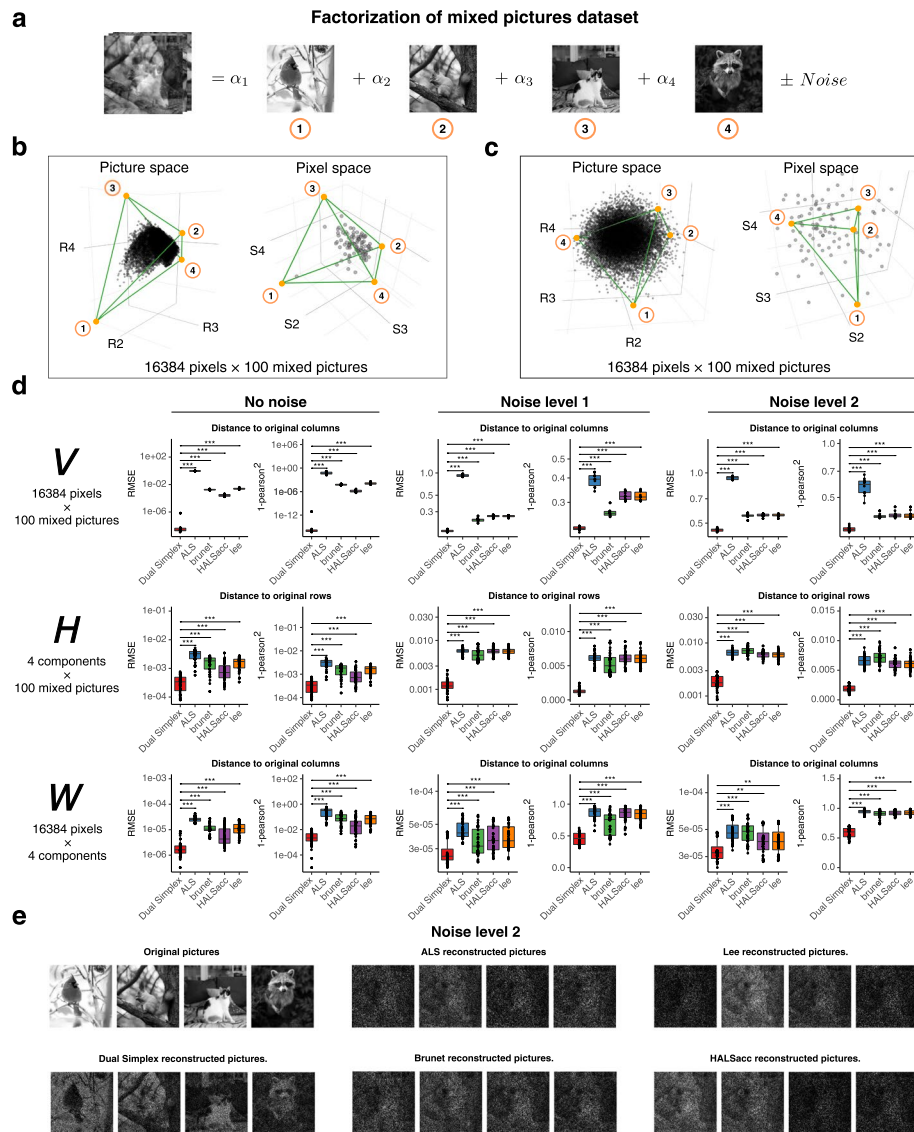


Fig. 4 Application of the Dual Simplex approach to the image data decomposition. **a** Each mixed image is obtained as a sum of the $K=4$ main picture components with nonnegative non-constrained coefficients sampled uniformly from the interval $[0,1]$. Noisy scenario involves the introduction of the proportional normally distributed noise with a predefined standard deviation representing the noise level. **b** Visualization of the individual pixel points (left) and picture points (right) in their respective projection spaces. Green lines represent true underlying simplex coordinates obtained by projecting rows of H and columns of W to respective spaces. For the no noise scenario, all the points are located within the geometrical simplex. **c** In case of proportional Gaussian noise present (original matrix multiplied by noise mask sampled from $N(1, \sigma_{\text{noise}}^2)$, $\sigma_{\text{noise}} = 2$ in this case) The simplex structure becomes distorted, and individual points can be located outside the original geometrical structure. **d** Reconstruction quality metrics measured for matrices V (row 1), H (row 2), and W (row 3) for different noise levels. Quality metrics are defined as median RMSE and $1-\text{pearson}^2$ values measured between rows of predicted matrix H and true matrix H , as well as columns of predicted matrices W and V in comparison to columns of the true matrices W and V . **e** Reconstructed images (columns of matrix W) visualized for all five methods for the high level of noise ($\sigma_{\text{noise}} = 2$)

perform its non-negative factorization, or to the special case of decomposition of the linearly mixed signals. Next, we will provide several examples of the practical applicability of this approach and discuss some special case implications of this approach for

problems such as image decomposition, supervised and unsupervised deconvolution of bulk gene expression data, and single-cell RNA-seq data analysis.

Application of the Dual Simplex approach to the NMF problem

To illustrate the applicability of our approach, we first applied our method to factorize a dataset, containing computationally mixed images (here, $K = 4$) with either no noise or random proportional noise added (Fig. 4a). Note that the images were mixed with random positive coefficients $\alpha_1, \dots, \alpha_4$ that were not required to sum to one. This setup represents a classical and straightforward NMF benchmark experiment, which allows both rigorous quantitative comparison between evaluated methods, and intuitive visual assessment of the recovered components (i.e., mixed images) [1]. The resulting mixed matrices were Sinkhorn-transformed, and the corresponding projections were visualized (Fig. 4b, c). The green lines on Fig. 4b and c represent the true simplex coordinates for either the picture or pixel space. It can be observed that in our image deconvolution scenario, even though the number of features (pixels) is high ($M = 16384$), pixel intensities in general are not specific to a particular picture, as all pixel points are relatively far away from the true vertices of a simplex structure (Fig. 4b, left). In case the number of mixed samples is significantly smaller than the number of pixels ($N = 100 \ll M = 16384$), the simplex of mixed images in the space of pixels will have only 100 points, which are also relatively far from the simplex vertices (Fig. 4b, right). Furthermore, the introduction of noise significantly distorts the structure of the data, making it even less related to the original simplexes for both picture and pixel spaces (Fig. 4c).

In direct comparison of the accuracy of V , H and W reconstruction, our method (Dual Simplex) outperforms commonly used NMF algorithms [28–31], yielding significantly more accurate matrices W and H for all noise levels, both in terms of RMSE error and Pearson correlation coefficient of rows/columns (Fig. 4d). We observe that since the number of features, describing the simplex in a picture space, is high ($M = 16384$), all NMF methods can reconstruct H with high accuracy for all scenarios. However, the most striking difference in performance becomes apparent for the reconstruction of the W matrix of the pure components, for which the utilization of the complementary simplex structure helps significantly. The major improvement delivered by the Dual Simplex approach is apparent from the comparisons based on both RMSE or correlation-based metrics, as well as directly by looking at the reconstructed images (Fig. 4e). Therefore, we conclude that the Dual Simplex approach is more robust to noise values and allows for partial reconstruction of the original pure picture components even in situations where other methods fail to do so. Additionally, we tested it with an increased number of mixed pictures in a dataset, where we observed the same pattern (Additional file 2: Fig. S3a, b).

Furthermore, to validate our approach in a most general and unconstrained matrix factorization task, we considered completely random positive matrices W and H ($M = 1000, N = 800$). For this scenario, the Dual Simplex approach again demonstrated improved accuracy of reconstruction for both clean and noisy data matrices W and H (Additional file 2: Fig. S3c). We also observed the same pattern that the matrix W is reconstructed significantly better by the Dual Simplex approach compared to other methods, especially in the presence of noise.

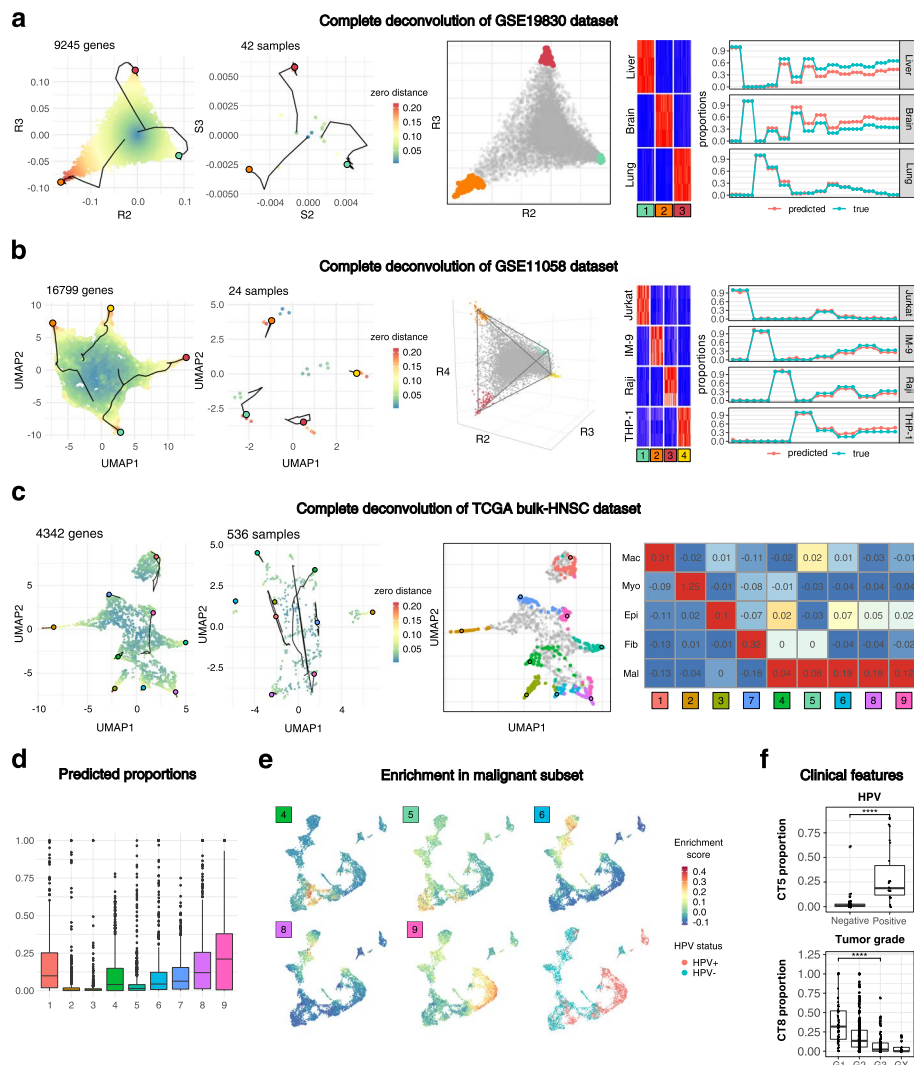


Fig. 5 Complete deconvolution of bulk-RNA datasets with the Dual Simplex approach. **a** Complete deconvolution of GSE19830 with projected data points colored by their distance from origin, final simplex vertices location and optimization history indicated by black lines for both gene and sample spaces (left), marker genes (obtained as genes with a highest relative log-fold change in the matrix W) are highlighted on a simplex of genes (middle) and expression of marker genes extracted from the pure samples (true matrix W) in the result reconstructed transcriptional profiles (computed matrix W) (heatmap). Predicted proportion values for each sample compared to true proportions extracted from the data (right). **b** Complete deconvolution of GSE11058 with projected data points colored by their distance from origin, final simplex vertices location and optimization history indicated by black lines for both gene and sample spaces (left), marker genes (obtained as genes with a highest relative log-fold change in the matrix W) are highlighted on a simplex of genes (middle), and expression of marker genes extracted from the pure samples (true matrix W) in the result reconstructed transcriptional profiles (computed matrix W) (heatmap). Predicted proportion values for each sample compared to true proportions extracted from the data (right). **c** Complete deconvolution of TCGA bulk-HNSC dataset with projected data points colored by their distance from origin, final simplex vertices location and optimization history indicated by black lines for both gene and sample spaces (left), marker genes (obtained as genes with a highest relative log-fold change in the matrix W) are highlighted on a simplex of genes (middle). Expression of predicted cell types' marker genes within the average cell type expression vectors extracted from the external GSE181919 single cell dataset (right). **d** Predicted proportions distribution for each of the main components. **e** Enrichment of predicted malignant cell subsets on a UMAP of malignant cells. Enrichment of the subset 5 correlates with HPV status coloring for the cells. **f** Correlation of predicted proportions with selected clinical features

Bulk gene expression deconvolution

Complete deconvolution

To demonstrate how the proposed NMF approach dissects real-world datasets of mixed samples, we first applied it to known benchmarks GSE19830 and GSE11058 [32, 33]. These datasets represent mixtures of 3 and 4 pure components, which yield underlying simplexes as triangle and tetrahedron-like structures (Fig. 5a, b). Search in the genes and samples spaces yield accurate vertices of the corresponding simplexes, which map to the genes specific to pure cell types and accurately recover proportions of pure components (Fig. 5a, b).

For a more complex and realistic scenario, we applied the Dual Simplex approach to a TCGA bulk RNA-seq dataset profiling $M = 548$ samples of the Head and Neck Squamous Cell Carcinoma (bulk-HNSCC) cancers characterized by global gene expression profiles. For the preprocessing of this type of data, we find it beneficial to keep only reasonably well-expressed genes and filter out noisy and housekeeping genes, which introduce distortion to singular value decomposition (see [Methods](#) for details).

We found our computational approach stably converging for the $K = 9$ cell type specific points perfectly aligning with the vertices of a gene simplex (Fig. 5c, S4a). To validate the results of the deconvolution, we assessed the enrichment profile of the predicted marker genes in the GSE181919 single-cell dataset of HNSCC [34]. Our analysis shows that each predicted cell type is biologically relevant to a specific subset of the single-cell data, revealing 4 major non-malignant cell types (including macrophages, myocytes, ciliated epithelial cells, and fibroblasts) as well as 5 different cancer subtypes (Fig. 5c, d). To understand if these cancer subtypes represent real subpopulations or mathematical artifacts, we zoomed into malignant cells and found that the five identified cancer subtypes are enriched across different malignant cell subclusters (Fig. 5e). To further confirm the biological relevance of the predicted signatures, we measured the enrichment of 100 markers per predicted cell type in MSigDB [35]. The enriched pathways for each cell type characterize the function of this cell type, which generally corresponds to the function of the single-cell population label assigned. (Additional file 2: Fig. S4b, c). Of interest, we found that cancer subtype 5 is significantly more abundant in the samples of HPV-positive patients and that cancer subtype 8 was inversely correlated with tumor grade (Fig. 5f). These experiments illustrate the applicability and effectiveness of our computational approach in solving the deconvolution problem for noisy simulated and high-dimensional real-world datasets.

Marker-based partial deconvolution

An important scenario for the deconvolution problem is its unsupervised case, when an algorithm solves the problem of dissecting the mixtures when only key marker genes of the pure components are available (e.g., spatial transcriptomics, scRNA-seq-based bulk deconvolution, etc.). Generally, knowing marker genes for a cell type is equivalent to knowing coordinates of the simplex vertices in a sample space (i.e., matrix W). The framework described in this paper provides a straightforward procedure for partial deconvolution using the Dual Simplex approach: the simplex of genes is now defined by an expression matrix V_{mark} , that contains only the selected marker genes. Deconvolution of V_{mark} leads to a proportion matrix H_{mark} . This reduces the problem to a conventional

partial deconvolution problem where the known proportions H_{mark} are used to compute profiles of the pure components, W , across all genes – both marker and non-marker genes (Methods).

We illustrate the marker-based partial deconvolution using two datasets: a single-cell derived computational mixing dataset by Sutton et al. [36] and GSE11058 data [33] of 4 mixed cell types (Additional file 2: Fig. S5). As can be seen, implementation of the Dual Simplex approach behaves robustly and accurately recovers proportions and gene expression profiles of pure components (Additional file 2: Fig. S5) compared to DSA [15] and CIBERSORT [37].

The DREAM challenge data deconvolution

To further validate and quantify the performance of the Dual Simplex, we compared it against state-of-the-art deconvolution algorithms in a real-world benchmark with known ground-truth. Specifically, we applied our method to the datasets from the DREAM deconvolution challenge [38]. This data comprises 8 different datasets containing both in-silico and in vitro mixtures of 15 pure cell-type populations prepared by different vendors (GSE199324).

We first identified a set of marker genes from the pure populations – gene expression profiles that were provided in the original publication but were not available to the original DREAM challenge participants – using a straightforward Limma-based marker selection procedure. These markers were used both for visualization in the reference-free deconvolution setting and as a basis for the solution in the marker-based solution (Fig. 6a). The extracted markers capture the main expression differences between the pure samples, as evident from both the marker expression matrix and the simplex representations generated with the Dual Simplex package (Fig. 6a).

Complete deconvolution

We applied the Dual Simplex approach to all eight datasets and compared its performance with leading reference-free methods: Linseed [21], TOAST [39], CellDistinguisher [40], and CAM3 [41], the latest version of the debCAM method [42]. Dual Simplex consistently achieved a marked improvement in the accuracy of cell type proportions recovery, as measured by Pearson correlation coefficients and the RMSE errors between predicted and true proportion vectors (Fig. 6b-c, Additional file 2: Fig. S6). The improvement was particularly pronounced for the in vitro mixtures (Fig. 6c bottom, Additional file 2: Fig. S6). Notably, two other gene simplex-based methods, CAM3 [42] and Linseed [21], ranked next in performance. This finding highlights that approaches leveraging the topological properties of the gene simplex in projected space, combined with effective denoising strategies, represent the most promising direction for advancing robust reference-free deconvolution algorithms.

Marker-based partial deconvolution

We then applied the Dual Simplex to the DREAM datasets in a marker-guided setting. Dual Simplex again outperformed all methods that participated in the challenge (Fig. 6d-e, Additional file 2: Fig. S6). We also compared the reference-free version of Dual Simplex in the same benchmark and found that it performs comparably to the best

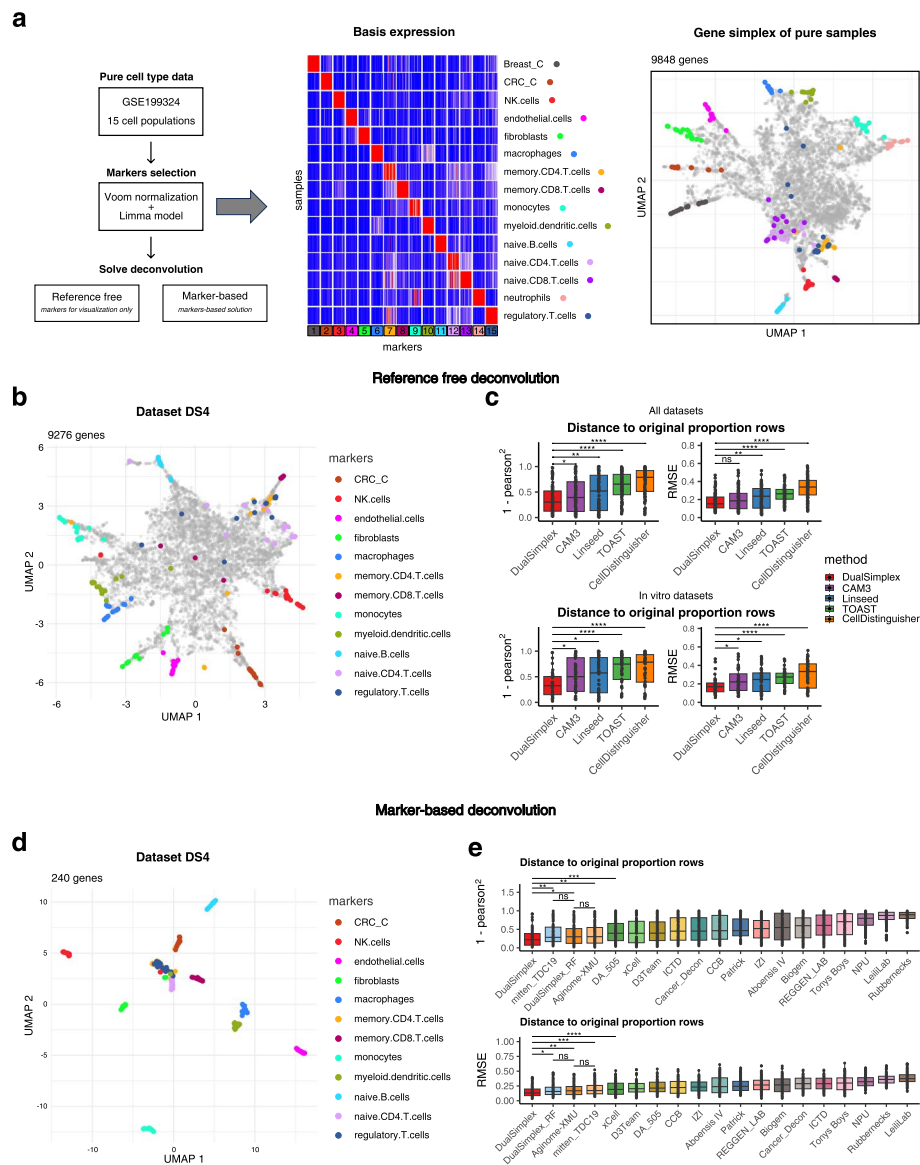


Fig. 6 The DREAM challenge competition data analysis. **a** Marker gene selection. Schematic overview of marker selection and its use in the experiment (left). Heatmap showing the expression of marker genes extracted from pure samples of the GSE199324 dataset within the corresponding average expression vectors (middle). UMAP representation of the gene simplex constructed using the Dual Simplex package from the matrix containing only pure samples of GSE199324, with the selected marker genes highlighted in color (right). **b** Reference-free deconvolution simplex. UMAP representation of the gene simplex constructed from the DS4 dataset using the Dual Simplex package, with the same marker genes highlighted in color. **c** Reference-free deconvolution results. Reconstruction quality metrics measured for matrix H calculated across all datasets (first row) and for in-vitro mixtures only (second row). Quality metrics are defined as median $1 - \text{pearson}^2$ and RMSE values measured between rows of predicted matrix H and true matrix H. **d** Marker-based deconvolution simplex. UMAP representation of the gene simplex constructed from the DS4 dataset using the Dual Simplex package, after retaining only the marker genes, with the marker genes highlighted in color. Marker genes extracted from pure cell type expression profiles, which were not available for original Challenge participants. **e** Marker-based deconvolution results. Reconstruction quality metrics measured for matrix H calculated across all datasets. Quality metrics are defined as median $1 - \text{pearson}^2$ (first row) and RMSE (second row) values measured between rows of predicted matrix H and true matrix H. Reference-free deconvolution results for the DualSimplex method were added for comparison and are labeled as DualSimplex_RF

marker-guided methods from the Challenge. This result demonstrates that the analytical characterization of the factorization problem underlying our approach translates into superior performance in both reference-free and marker-based deconvolution scenarios.

Single-cell data analysis

Another interesting direction of the application of our method is a single-cell RNA-seq data analysis, where each sample typically corresponds to a pure cell type (aside from the exclusion of doublets and other artifacts). Conceptually, non-negative factorization of such data should recover the underlying clusters [31], making it a deconvolution problem with only pure components present.

To demonstrate the ability of our method to solve this task, we applied the method to a GSE103322 single-cell dataset of head and neck squamous cell carcinoma, containing 5902 cells extracted from 18 cancer patients. This dataset contains 2215 malignant as well as 3363 non-malignant cells of primary oral cavity tumors, as well as metastatic lymph nodes. Additional file 2: Fig. S7a reproduces the t-SNE plot from the original paper [43] with annotation provided by the authors. After retaining the 3000 most variable genes [44], filtering outliers, and applying the Sinkhorn transformation, we projected the data into SVD space. Both cell and gene projection revealed clear subgroup structure (Additional file 2: Fig. S7b). UMAP visualization emphasized cell-type-specific genes at simplex vertices (Additional file 2: Fig. S7c), and the optimization procedure aligned the solution around these vertices. The resulting proportion matrix H defined cell clusters, and the columns of W corresponded to transcriptional profiles matching known subsets reported by the original study (Additional file 2: Fig. S7d-f).

The Dual Simplex approach provides an intuitive procedure and direct control over the clustering process. Indeed, one of the aspects of the commonly used single-cell clustering pipelines is the need to build the neighborhood graph to understand the spatial distribution of the points within the space of main variational components, followed by clustering based on the distance thresholds. Such an approach results in an unpredictable number of clusters. Our approach, on the other hand, performs simultaneous alignment and clustering of the cells and genes, providing a natural way to perform clustering of the data with an explicitly controllable number of clusters (K) (Additional file 2: Fig. S7g).

The Dual Simplex approach advantages

The key advantage of Dual Simplex is its ability to jointly exploit both feature and sample spaces, rather than relying on only one. Modern topological mixed-matrix analysis algorithms operating in projected spaces, such as CAM3 [42] and Linseed [21], restrict their analysis to a single space. Therefore, they require a sufficient number of component-specific features or high-proportion samples. By contrast, our approach leverages the fact that the input data inherently encodes *two* simplex structures: (1) a simplex of features revealed by row normalization (i.e., in the sample space) and (2) a simplex of samples revealed by column normalization (i.e., in the feature space).

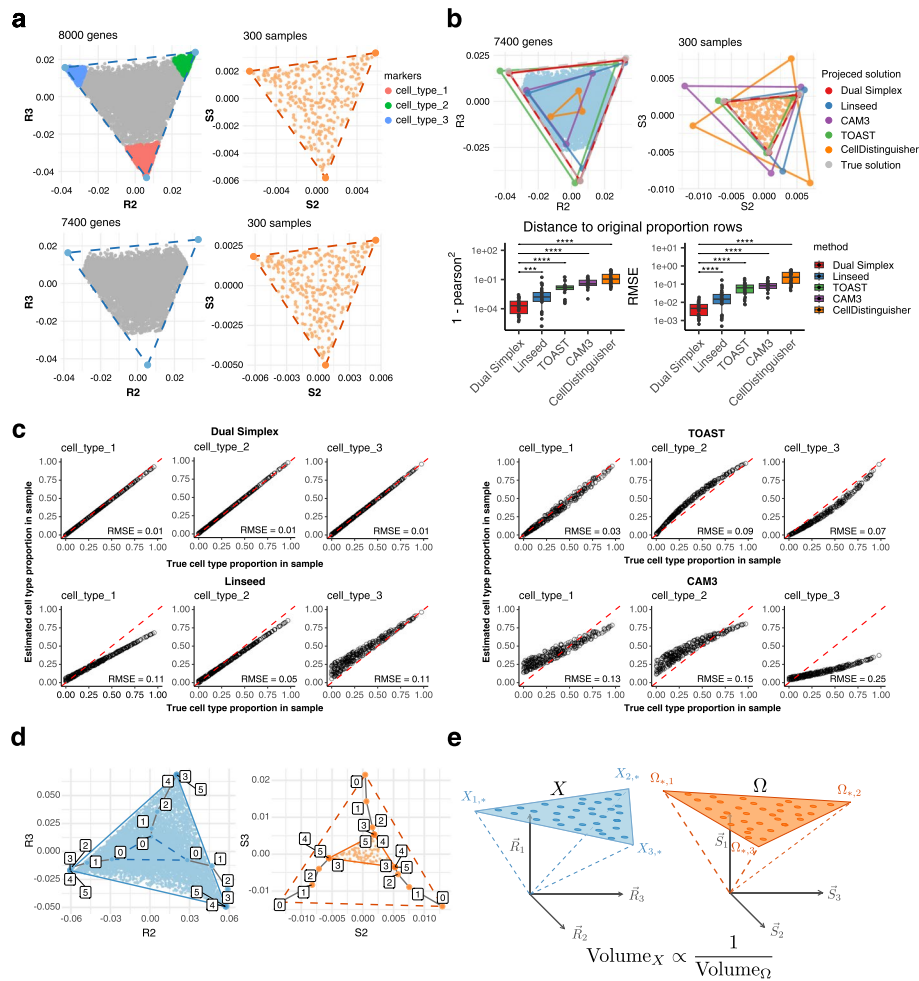


Fig. 7 The Dual Simplex approach advantages. **a** Missing component-specific feature scenario. Genes and samples simplexes in their respective projection spaces for simulated gene expression data ($N = 8000$, $M = 300$, $K = 3$, $\sigma_{\text{noise}} = 1.5$) using the Dual Simplex package. The first row visualizes the dataset including marker genes, which were selected as the top 200 genes with the highest log-fold change for each cell type (600 genes in total) and highlighted in color. The second row visualizes the same dataset after removing these marker genes. True simplex vertices are indicated with dashed lines. **b** Performance of the algorithms in case missing component-specific features. The first row shows genes and samples simplexes in their respective projection spaces for one of the simulated matrices. Vertices for simplexes (obtained by applying the same Sinkhorn transformation as was applied to the original matrix V to the solution matrices W and H for each individual method) are indicated with colored lines. True solution vertices are indicated with gray dashed line. The second row indicates the reconstruction quality metrics measured for matrix H calculated of 10 independently simulated matrices. Quality metrics are defined as median $1 - \text{pearson}^2$ and RMSE values measured between rows of predicted matrix H and true matrix H . **c** Proportion values predicted by the top four scoring methods for a single simulated matrix. Each dot represents the relationship between the true proportion (x-axis) and the predicted proportion (y-axis) for a given sample, estimated by the corresponding method for each cell type. **d** Gradient descent optimization of the simplex vertices using the minimal formulation of the Dual Simplex problem, which involves searching for a single vertices matrix under a constraint on its inverse, applied to simulated gene expression data. ($N = 10000$, $M = 100$, $K = 3$, σ_{noise}). **e** Geometrical interpretation of original simplexes in features and samples spaces highlights inverse proportionality of their volumes

Conceptually, this duality reflects the equivalence of features and samples in the linear unmixing problem. Importantly, if the simplex structure in one space is distorted – due to noise, missing features, or weakly informative variables (e.g., pixels in image

data that do not differ strongly across images (Fig. 3)) – our algorithm can rely on the complementary simplex in the other space to recover the solution.

We illustrate this advantage using simulated gene expression data in which markers of the main pure sample profiles were deliberately removed (Fig. 7a). In this setting, other deconvolution methods fail to accurately recover the true proportions. By contrast, Dual Simplex successfully reconstructs the proportion matrix with high accuracy, demonstrating superior robustness in challenging scenarios (Fig. 7b-c).

In fact, the geometrical relation between simplexes can be further characterized. By making a simple substitution in the joint optimization problem formulation (Eq. 9), namely $\tilde{\Omega} = \sqrt{\Sigma_{ss}} \times \Omega \times \sqrt{D}$ and $\tilde{X} = \sqrt{D} \times X \times \sqrt{\Sigma_{ss}}$, we obtain the following reformulated optimization problem to its minimal form:

$$\begin{aligned} \tilde{X}, \tilde{\Omega} = \operatorname{argmin} \quad & \|\tilde{\Omega} \times \tilde{X} - \mathbb{I}_{K \times K}\| \\ & \tilde{X} \quad \text{s. t. } S^T \times \sqrt{\Sigma_{ss}} \times \tilde{\Omega} > 0 \\ & \tilde{\Omega} \quad \tilde{X} \times \sqrt{\Sigma_{ss}} \times R > 0 \\ & \sqrt{N}\tilde{X}_{*,1} = \sqrt{M}\tilde{\Omega}_{1,*} \end{aligned} \quad (10)$$

In this formulation (Eq. 10), the matrices $\tilde{X}$ and $\tilde{\Omega}$ will now represent modified simplex coordinates in projected space, which are mutual inverses of each other (Additional file 1: Supplementary Note 10). This reduces the optimization to the problem of estimating a single simplex coordinate matrix, with specific positivity constraints on it and its inverse. As a theoretical aside of this work, we illustrate the concept of such algorithms using simulated data (Fig. 7d).

Notably, in this formulation, the size of the gene simplex is inversely proportional to the size of the sample simplex (i.e., a small simplex in one space corresponds to a large simplex in another). We further proved this proportional relation analytically and showed that it extends naturally to the volumes of the original simplex coordinates X and Ω (Fig. 7e, Additional file 1: Supplementary Note 10). This observation suggests that solving the NMF problem can fundamentally be viewed as identifying minimal-volume simplex structures enclosing points in both the sample and feature spaces, requiring the optimization of only a single coordinate matrix.

Finally, this reformulation also allows for a substantial reduction in the search space. For instance, consider a dataset with 500 samples (N) and 10,000 features (M), where mixtures are composed of only three main components. In the conventional NMF setting, the optimization would involve estimating 31,500 parameters (30,000 elements of W and 1,500 elements of H). In contrast, our approach reduces the task to identifying only 6 unknown parameters – corresponding to the free elements of X (more generally $K(K-1)$ variables). This dramatic dimensionality reduction highlights not only a theoretical advantage but also a practical one, enabling scalable deconvolution of large-scale datasets.

Non-uniqueness of the NMF solution

As noted previously, NMF solutions are generally non-unique: matrices W and H can be obtained in multiple forms, especially when the input matrix V is noisy or lacks sufficient constraints [24]. Consequently, the Dual Simplex solution is also not unique.

To illustrate, we first generated a simple mixed matrix, containing only 6 features and 6 samples, previously shown to admit multiple factorizations [24] (Additional file 2: Fig S8a). As expected, Dual Simplex recovered two distinct sets of vertices consistent with these solutions (Additional file 2: Fig. S8b). In practice, when the factorized matrix lacks a clear simplex structure – e.g., due to missing pure samples or cell-type-specific features – the Sinkhorn projections become unorganized, which we demonstrate using simulations (Additional file 2: Fig. S8c). In such cases, infinitely many simplex vertex configurations can enclose the data points and still reconstruct V precisely. Accordingly, Dual Simplex also yields multiple solutions, depending on initialization (Additional file 2: Fig. S8d). However, both simplexes are connected, have minimal volume, and enclose all the points in both spaces. As a consequence, these simplexes produce a valid solution. This result would be harder to achieve with this kind of data if only one of the simplexes were analyzed, as demonstrated in Fig. 7.

Discussion

Our work builds on the idea that the problem of linear and sum-to-one constrained matrix unmixing is, in essence, a problem of simplex identification [45]. Various approaches for finding such simplexes were proposed for general hyperspectral images [45–47]. With the introduction of the non-negative matrix factorization problem [1], the linear unmixing was recognized to be a sum-to-one constrained NMF problem [48]. Furthermore, it was noted that the general NMF problem can be transformed into a linear unmixing/deconvolution problem via matrix normalization and then solved by a convex analysis framework [23]. In biological settings, we and others explored simplex-based methods in the context of gene expression data deconvolution [21, 22, 49].

The key distinction of the Dual Simplex approach compared to other simplex-based or convex algorithms [19–23, 41, 49] is in introduction and utilization of two connected simplex structures within an analytically defined projected space. CAMNS [23] and ParTI [49] operate on the simplex of samples in feature space and may fail when high-proportion components are missing. Variants of CAM [20, 22, 41], Linseed [21], and SISAL [19] instead analyze the simplex of features in sample space, which can lose descriptive power when component-specific features are missing. Ultimately, all these methods attempt to identify simplex vertices in one space [23], but they differ in how this is achieved. CAMNS identifies points on the simplex boundary through random search of directions, assuming that solution points always lie on the boundary. This condition is not guaranteed under noise, as demonstrated by Linseed and in our Fig. 3. ParTI and Linseed rely on SISAL for vertices detection. While SISAL directly optimizes simplex vertices and can tolerate noise, it is formulated for a single simplex and cannot directly capture the joint structure of two related simplexes. CAM defines simplex vertices as a location of component-specific feature clusters that might be missing in the data.

In contrast, by using the projective formulation of the Sinkhorn-transformed NMF problem [8], the Dual Simplex explicitly links the feature-space simplex with the sample-space simplex. Its optimization problem incorporates hinge function-like terms similar to SISAL but is defined on Sinkhorn-transformed matrices and operates simultaneously on both simplex structures in SVD space. The resulting loss function combines convex

components, which guarantees the convexity of the overall optimization. In essence, the Dual Simplex approach balances two complementary objectives, each acting on the respective simplex coordinates. This structure allows a broad range of optimization strategies to be applied within our framework, including the Lagrangian-based approach proposed in the SISAL method [19] which would be the subject of subsequent work.

Our approach yields a unified Dual Simplex optimization formulation of the NMF/deconvolution problem (Eq. 9, Fig. 3). It reveals that the NMF problem can be effectively reduced to optimization of $K(K - 1)$ independent variables, with K being the number of cell types, clusters, or independent components in general.

The Dual Simplex approach provides unique power in terms of dimensionality reduction and robustness relative to imbalance between the number of features and samples, as optimization requires the best fit for both simplexes. Indeed, our results from unconstrained nonnegative factorization of image datasets demonstrate that optimization of both simplexes makes our algorithm more robust to noise and incomplete data and provides significantly more accurate decomposition matrices for the general NMF problem compared to other methods, even though the optimization itself is conducted in the space of much lower dimensions (Fig. 4, S3).

Moreover, our approach provides a unified framework for solving different flavors of the sum-to-one constrained version of the NMF problem, i.e., the deconvolution problem, which is commonly utilized in computational biology due to the wealth of information from RNA-sequencing data. These flavors include complete and marker-based deconvolution of bulk-RNA sequencing data (Figs. 5, 6, Additional file 2: Fig. S4, S5, S6). Furthermore, one can theoretically expand this approach to address the *incomplete deconvolution* problem, i.e., the situation when some of the cell types in the mixture are known while others are not. In this problem, one can freeze the solution in simplex vertices for already known markers (e.g., using the marker-based approach described), and then perform optimization, searching for the rest of the vertices in a low-dimensional space, utilizing the geometrical properties of the described simplex structure.

Besides that, we demonstrate that our method can efficiently dissect the cluster structure of the data, e.g., single-cell RNA-seq data (Additional file 2: Fig. S7). From one perspective, this problem can be regarded as an extreme case of deconvolution with only pure components present in the data. Therefore, in the feature (genes) space, sample dots are concentrated in the vertices of the simplex with almost no points in between. In these settings, deconvolution of the scRNA-seq data would be equivalent to clustering of the cells in the space of genes. On the other hand, one can think about obtained pure components in terms of archetypes constituting the expression profile of a single point, as described by Alon and colleagues for bulk and single-cell gene expression data [50]. This idea was then developed using different computational and numerical approaches, such as kernel analysis, etc. [51, 52]. Our work effectively provides a rigorous mathematical basis and procedure for the implementation of the archetypal (i.e., simplex) structures within the gene expression data.

More broadly, the ability to solve the NMF problem suggests the applicability of our approach to a wider range of tasks, such as those that can be formulated as NMF problems, including spatial transcriptomics analysis [4], mass spectrometry analysis [5], and spectral cytometry analysis [6], as well as applications beyond computational biology.

Conclusions

We developed a novel analytical framework for non-negative matrix factorization (NMF) grounded in the topological properties of the problem, connecting two simplexes through the Sinkhorn normalization procedure in a well-defined projection space. The described Dual Simplex approach reduces the dimensionality of the problem, making the optimization process more robust and accurate, particularly in noisy data settings. The framework is versatile, addressing different versions of the sum-to-one constrained NMF problem common in computational biology, such as transcriptional data deconvolution and single-cell RNA-seq data analysis. This method provides a unified and efficient approach to deconvolving complex biological data by leveraging the geometrical properties of simplex structures.

Methods

Data and implementation

The entire framework is implemented using the R language with a C++ backend and the RcppArmadillo package used for linear algebra calculations [53].

The package available online in the form of a source code repository (<https://github.com/artiomovlab/dualsimplex>). Scripts to reproduce all the figures, as well as corresponding data, are stored in a separate repository (https://github.com/artiomovlab/dualsimplex_paper).

Simulation data generation

For general matrix unmixing problems $V = W \times H$, matrices H and/or W were sampled from a random uniform distribution $W, H \sim U(0, 10)$. The additive proportional noise was then applied using the formula $V_{\text{noisy}} = V * (1 + N(0, \sigma_{\text{noise}}^2))$, where σ_{noise} represents the chosen level of noise deviation. Negative elements of such a matrix were set to zero.

For gene expression scenarios, when generating basis vectors (columns of the matrix W), expression values x were chosen using a formula $x = 2^X$ where X is a random variable sampled uniformly from two normal distributions, specifically $\mathcal{N}(4, 0.75)$ and $\mathcal{N}(10, 1.5)$, with a 2/3 ratio. This type of expression generation was designed with the assumption of a bimodal nature of the expression data. The first mode represents the expression of lowly or non-expressed genes, while the second mode represents well-expressed genes. Conversely, the proportions for a given cell type (rows of the matrix H) were sampled as random points within a geometrical simplex structure to enforce the sum-to-one constraint. In the case of a noisy data scenario, the noise values (y) were generated using a formula $y = 2^{\sigma_{\text{noise}} Y}$ where Y follows standard normal distribution ($\mathcal{N}(0, 1)$) and σ_{noise} represents the chosen level of noise deviation.

Gene expression data preprocessing

To apply our method to a real expression dataset, we recommend a three-step procedure for data cleaning, which enhances the accuracy of identification for the hyperplanes of the simplexes (i.e., vectors R and S).

Step 1: Data prefiltering. We suggest prefiltering highly correlated genes, such as ribosomal and mitochondrial genes. Additionally, it is beneficial to remove housekeeping genes since such genes are not related to any cell type specifically, but can introduce bias during singular value decomposition. For improved biological interpretation, we also remove non-coding genes, although it is not necessary from the algorithmic perspective. Furthermore, since an exact definition of housekeeping genes is lacking, we found it practical to not only remove genes derived from previously designated housekeeping gene lists but also to eliminate those with expression profiles that highly correlate with these genes. To achieve this, we employed a straightforward KNN-based approach [54], considering the proximity between values of genes in the projected space, as this proximity reflects the similarity in their expression patterns across samples. Complete lists of genes specific to each of the filtered subsets could be found in (Additional file 3: Table S1).

Step 2. Filtering based on variability. Since our approach leverages linear components of variability in gene expression to dissect cell type composition, it is beneficial to remove the genes that are lowly varying. To achieve this, we perform median absolute deviation (MAD) filtering, keeping genes that have MAD greater than a selected threshold. The precise number of genes filtered was chosen based on each dataset specifically. The distribution of MAD in the data typically forms a distinctive bimodal distribution, where one group of genes contains lowly varied genes and technical artifacts, whereas the second group of genes contains highly varied genes. The threshold for the filtering was chosen for the first group to be filtered out.

Step 3: Data denoising/Outlier removal. Another way to improve the projection optimization solution is to denoise the data by removing outlier genes and samples, since SVD-based approaches are generally sensitive to outliers. Here, we suggest using the approach of removing genes/samples that are too far from the projection hyperplane in terms of geometrical distance. For each gene or sample point, we calculated two values. The first is the zero distance, defined as the Euclidean distance between the point and the mean value in the respective projected space (i.e., its position within the coordinate system $R_2, R_3, \dots, R_K$ or $S_2, S_3, \dots, S_K$). The second is the plane distance, which quantifies the variance of the gene (or sample) vector not captured by the K projected vectors. It is computed as the Euclidean distance between the respective row (or column) of the residual matrix $(V_{ss}^\infty - S \times V_{ss}^\infty \times R^T)$ and the corresponding mean vector. Decisions regarding gene or sample removal are made based on these values.

Our approach implements an iterative filtering process where step 3 can be repeated, allowing for gradual strengthening of filtering conditions. The improvement is generally evaluated via 2-dimensional projection plots to major singular vectors, as well as dot plots of plane and zero distances, to observe a reasonably homogeneous group of dots without distant outliers (see Additional file 1: Supplementary Note 12 for details).

Optimization procedure

Minimization problem (Eq. 9) was solved using a gradient descent optimization procedure (see Additional file 1: Supplementary Note 10 for details). The main

implementation uses the cost function (Fig. 3d) and is, in essence, a block-wise gradient descent procedure operating in a projection space with respect to 3 matrix variables (X, Ω, D) with the sequence of updates $X \rightarrow D \rightarrow \Omega \rightarrow D$. For the variables X and Ω the algorithm computes partial derivatives ∇F_X and ∇F_Ω which are then used in update rules $X_{i+1} = X_i - \mu * \tilde{R}_X * \frac{\nabla F_{X_i}}{\|\nabla F_{X_i}\|_2}$ and $\Omega_{i+1} = \Omega_i - \nu * \tilde{R}_\Omega * \frac{\nabla F_{\Omega_i}}{\|\nabla F_{\Omega_i}\|_2}$. Here $\mu * \tilde{R}_X$ and $\nu * \tilde{R}_\Omega$ are scaling constants for step magnitude based on the average norm of projected vectors in their respective spaces. New values for the variable D are calculated analytically via NNLS:

$$\text{vec}(D) = \underset{\vec{D}}{\text{argmin}} \left\| S \times \overset{\infty}{V}_{ss} \times R^T - \sum_{j=1}^K d_j \times \begin{bmatrix} \omega_{1,j} \\ \vdots \\ \omega_{K,j} \end{bmatrix} \times [x_{j,1} \cdots x_{j,K}] \right\|_F^2.$$

Initialization strategies

To select initial values for matrices X and Ω , we employed three different strategies: *random centered*, *select_x*, and *select_omega*. The first strategy involves selecting vertex points that are centered around zero for a set of randomly generated coordinates.

$$X_{\text{init}} = \underset{X_1, X_2, \dots, X_L}{\text{argmin}} \sqrt{\sum_{i=1}^K \left(\frac{\sum_{j=1}^K x_{j,i}}{K} \right)^2} \quad \Omega_{\text{init}} = \underset{\Omega_1, \Omega_2, \dots, \Omega_L}{\text{argmin}} \sqrt{\sum_{i=1}^K \left(\frac{\sum_{j=1}^K \omega_{i,j}}{K} \right)^2}.$$

The matrix D for this scenario is obtained as a solution for the non-negative least squares (NNLS) problem:

$$\vec{D} = \text{NNLS}(Q(X, \Omega), \text{vec}(S \times \overset{\infty}{V}_{ss} \times R^T)).$$

Where $Q(X, \Omega)$ is a $K^2 \times K$ matrix. Each column of Q is computed as the vectorized product of the corresponding column of Ω and row of X .

$$\forall j \in [1, K] Q_{*,j} = \text{vec} \left(\begin{bmatrix} \omega_{1,j} \\ \vdots \\ \omega_{K,j} \end{bmatrix} \times [x_{j,1} \cdots x_{j,K}] \right).$$

The *select_x* and *select_omega* initialization methods involve geometrical initialization using the VCA algorithm [47] for one of the initial simplexes (e.g. X), followed by analytical imputation of another one (e.g. Ω). This scenario is especially effective in cases when one of the simplexes is described better due to differences in the number of points (e.g. $N \gg M$) or the quality of these points (e.g., for single-cell data, we have only pure component points for Ω , which already describes the simplex).

We started with geometrical initialization of the first simplex:

$$X_{\text{init}} = \text{VCA}(\overset{\infty}{V}_{ss}^T) \text{ or } \Omega_{\text{init}} = \text{VCA}(\overset{\infty}{V}_{fs})$$

After this, we get the $\vec{D}$ based on the known values using Sinkhorn transformed matrices properties, which we have proven (Additional file 1: Supplementary Note 10).

$$\vec{D} = \frac{M}{N} X^{-T} \times \begin{bmatrix} \sum_{j=1}^N r_{1,j} \\ \sum_{j=1}^N r_{2,j} \\ \vdots \\ \sum_{j=1}^N r_{K,j} \end{bmatrix} \quad \text{or} \quad \vec{D} = \Omega^{-1} \times \begin{bmatrix} \sum_{i=1}^M s_{1,i} \\ \sum_{i=1}^M s_{2,i} \\ \vdots \\ \sum_{i=1}^M s_{K,i} \end{bmatrix}.$$

And finally obtain missing coordinates for the second simplex.

$$\Omega_{\text{init}} = S \times \bar{V}_{ss}^\infty \times R^T \times \left(\text{diag}(\vec{D}) \times X \right)^{-1} \quad \text{or} \quad X_{\text{init}} = \left(\Omega \times \text{diag}(\vec{D}) \right)^{-1} \times S \times \bar{V}_{ss}^\infty \times R^T.$$

Hyperparameter tuning

The algorithm has four main parameters: μ , ν , λ , and β . While in some tasks a single parameter setting is sufficient, solving NMF problems often requires higher precision, and thus multiple optimization runs with different values are necessary. We found the following strategy useful for tuning the parameters of the Dual Simplex optimization.

Parameters μ and ν control the learning rates of the respective coordinates X and Ω . Smaller values yield smaller coordinate updates. We recommend starting with $\mu = \nu = 1$ (full gradient step) and gradually reducing the rates to refine the solution. In practice, we run the algorithm for several iterations (e.g., 1000), reduce μ and ν by a factor of 10, repeat optimization, and continue this schedule until convergence.

Parameters β and λ regulate the weighting of the loss function components (Fig. 3c), which include: (1) deconvolution error, and (2) negativity penalties. The deconvolution error measures consistency between simplex coordinates, effectively determining simplex size. Negativity penalties enforce that the solution lies within the positive span of the estimated W and H matrices [24]. Ideally, the contributions of these error terms should be balanced so that both the deconvolution and negativity terms converge to comparable magnitudes.

To achieve this, we adopt a scheduling strategy in which negativity penalties are gradually increased during optimization. Geometrically, this works as follows: when the deconvolution term dominates, the algorithm inflates the simplex to enclose all data points. As the penalty terms grow, the vertices are rotated toward the positive span, pushing them closer to the true boundary of the data. Simultaneously, μ and ν are decreased step by step (e.g., across 12 stages from $\mu = \nu = 1$ down to $\mu = \nu = 10^{-12}$). At each stage, the negativity penalty coefficients are amplified multiple times according to $\alpha = \mu^2 \cdot 100^i$ and $\beta = \nu^2 \cdot 100^i$, where i is the iteration index (e.g., 5 times over 1000 iterations).

This scheduling procedure is implemented as the default optimization method in our package. It was designed to ensure that all cost function terms converge smoothly and remain comparable in magnitude (see Additional file 1: Supplementary Note 10 for further details).

Non-negative matrix factorization

For the NMF problem statement, we consider $K = 4$ different flattened pictures of the same size ($M = 16384$ pixels), which are utilized as columns of the original

components matrix (16384 pixels $\times$ 4 pictures). The input matrix V comprises the chosen number N of mixed pictures (for example $N = 100$ for Fig. 4), where each column i was generated as a sum of main component pictures with different coefficients:

$$V_{*,i} = \sum_{k=1}^K \alpha_{k,i} W_{*,k}.$$

Here, coefficients $\alpha_{k,i}$ constitute the columns of the coefficient matrix H . These are sampled uniformly from the interval $[0,1]$ and do not necessarily sum to one. In the case of noisy matrices, we generated them as $V_{\text{noisy}} = V * (1 + N(0, \sigma_{\text{noise}}^2))$, where σ_{noise} represents the chosen level of noise deviation.

The Dual Simplex procedure was applied to the result matrix V_{noisy} without additional filtering steps. The optimization procedure was randomly initialized, and optimization was performed according to the default optimization strategy described above.

Additionally, for each scenario four different NMF algorithms Lee [29], Brunet [28], HALSacc [30] and ALS [31], were tested using the NMF R package [55].

For each NMF experiment, we tested all methods using 10 independently generated noisy matrices V_{noisy} . Since some NMF algorithms can be unstable, we performed 10 independent training runs for each method and selected the best solution using the NMF R package. The total number of runs is therefore (number of matrices) $\times$ (number of methods) $\times$ (number of runs), with (number of matrices) $\times$ (number of methods) solutions selected for the plots.

Comparison of different NMF methods

Since nonnegative matrix factorization does not have any special constraints for W and H , these matrices could only be reconstructed up to the scaling factors of the columns of W and rows of H . This produces incomparable matrices W and H for different methods. To make them comparable, we scale the columns of W and rows of H , representing all solutions in terms of column-normalized matrix W_{colnorm} , row-normalized matrix H_{rownorm} and the scaling matrix D .

$$V = W \times H \Leftrightarrow V = W_{\text{colnorm}} \times \underbrace{D_w^{-1} \times D_h^{-1}}_D \times H_{\text{rownorm}}.$$

The accuracy of such unified NMF solutions was estimated using two metrics: the RMSE error (rmse) and transformed Pearson correlation between matrix values (pearson). Normalized columns of the solution basis matrix W'_{colnorm} were compared to the normalized columns of the original matrix W_{colnorm} , and vice versa, normalized rows of the solution matrix H'_{rownorm} were compared to the original normalized rows of the matrix H_{rownorm} . For matrix V the same functions were measured for the columns of the matrix.

$$\begin{aligned} &\text{for the column (row) } X \text{ of the matrix } W \text{ and } V (H) \\ \text{rmse}(X) &= \frac{\sum (x_{\text{true}i} - x_i)^2}{\text{length}(X)} \quad \text{pearson}(X) = 1 - \left(\frac{\text{cov}(X_{\text{true}}, X)}{\sigma_X \sigma_{X_{\text{true}}}} \right)^2. \end{aligned}$$

TCGA bulk-HNSC dataset deconvolution

The dataset was downloaded using TCGAbiolinks R package [56, 57]. The clinical data, was retrieved directly from the GDC Data Portal [58], using the project_id TCGA-HNSC.

The dataset was filtered according to the following proposed procedure. First, we applied a MAD cutoff by the threshold of $\log_{mad} = 0.1$. After that, we calculated an initial projection of the data points onto simplex hyperplanes ($K = 20$) and removed house keeping genes with their closest neighbors in the genes hyperplane, using the threshold of $KNNd = 0.015$. For outlier filtering, we employed consequent distance cut-offs, reaching thresholds of $zero_distance = 0.2$, $plane_distance = 0.15$ for genes and $zero_distance = 0.1$, $plane_distance = 0.045$ for samples (See Additional file 3: Table S2 for more details). This resulted in a dataset with $M = 4326$ genes and $N = 535$ samples. The information about removed and retained genes is available in Additional file 3: Table S1.

Housekeeping and non-coding gene annotations were retrieved from publicly available sources [59, 60] and Ensembl [61] respectively.

After this initial filtering, the elbow point in the SVD plot began to manifest, establishing a range of approximately 7 to 12 potential cell types for analysis. For cell type numbers within this range, hyperplane projections were calculated. To determine the final number of cell types, the deconvolution procedure was executed across this range of 7–12 potential cell types. Several criteria were used to identify the optimal number of cell types: the alignment between the number of vertices on the genes UMAP and the number of points in the solution; the distinctiveness and specificity of the predicted cell type signatures; and the biological relevance of the identified cell types. Using these criteria, it was established (subjectively) that the most suitable number of cell types for demonstration purposes is 9. It is noteworthy that the deconvolution can be extended to accommodate a larger number of cell types, unveiling smaller and less distinct cell subpopulations. However, such subpopulations, although genuine, were not as suitable for demonstration with a single-cell reference.

After that, a deconvolution, using optimization, was performed. The gradient descent was initialized using the select_x method and run for 9000 steps, with $\lambda = 0.01$, $\beta = 0.01$, and learning rates (μ and ν) of 0.001.

TCGA bulk-HNSC deconvolution results analysis

Marker gene enrichment in single-cell data was calculated by taking $n = 100$ top marker genes per cell type and calculating module scores for them using the AddModuleScore function from Seurat 4.0.0 [44]. Aggregated heatmaps show the average expression across all cells within each analyzed single-cell cluster.

Pathway analysis was performed by taking the same marker genes and calculating p-values for their enrichment in reference gene sets with clusterProfiler 3.18.1 [62]. MSigDB gene sets from the H, C2:CGP, GO:BP and GO:CC categories were retrieved using msigdb 7.2.1 [63] and used as a reference for clusterProfiler.

For clinical associations analysis (Fig. 5f), sample annotation was obtained from the GDC portal [58]. Potentially relevant features were selected manually, and numerical features were converted to binned categorical values. Subsequently, Kruskal–Wallis

tests were conducted for each of the selected features, considering each cell type. The objective of the experiment was to assess whether samples, separated based on specific feature values, demonstrate statistically significant differences in predicted cell type proportions.

GSE181919 single-cell data preprocessing

The whole dataset, without additional filtering, as well as the malignant subset, were separately scaled and normalized using the standard Seurat v. 4.0.0 procedure, integrated using Harmony 1.2.0 [64] with parameter $N_PC = 20$, and mapped onto UMAP coordinates, in which form they are presented in Fig. 5.

Marker-based deconvolution algorithm implementation

The algorithm for marker-based deconvolution (Additional file 2: Fig. S5) accepts the same gene expression matrix V as the complete algorithm. It also requires a list of marker gene names for each of the cell types present in the mixture.

The expression matrix and marker genes for a deconvolution benchmark GSE11058 dataset were obtained using the CellMix package [14]. The complete expression matrix comprises 24 samples of 4 mixed cell types with expression values for 7862 genes. For each cell type, 20 marker genes were selected as input for the algorithm, resulting in a reduced expression matrix V_{mark} (80 genes $\times$ 24 samples). Solution for each cell type was obtained during optimization as if we solved a reference-free deconvolution problem. This initialization produced a signature proportion matrix H_{mark} (4 cell types $\times$ 24 samples). This matrix was then used to infer the basis matrix (W (7862 genes $\times$ 4 cell types)) through the conventional NNLS procedure using the original matrix V :

$$W = \underset{W'}{\operatorname{argmin}} \|V - W' \times H_{\text{mark}}\|_F^2.$$

The same procedure was applied to deconvolve a single-cell-based simulation dataset [36], consisting of $K = 7$ cell types mixed in $N = 100$ samples. For this data, three different marker sets were extracted from [36], specifically VL [65], NG [66], CA [67]

This approach can be particularly useful when applied to the datasets where the simplex structure is distorted due to technical or biological noise, making precise detection of the simplex plane through SVD and simplex vertices through optimization a challenging task. In these situations, the removal of a significant portion of the genes effectively eliminates unwanted variations associated with those genes, while still retaining the essential vertices of the gene simplex.

DREAM challenge data deconvolution

To prepare marker sets for deconvolution, we extracted pure cell type populations from the GSE199324 dataset. For simplicity, we derived markers using the pure cell-type expression profiles provided in the original publication. We acknowledge that these pure profiles – and therefore these specific markers – were not available to the original DREAM challenge participants. Thus, the comparison in Fig. 6e is not fully equivalent. Importantly, these expression signatures were not used elsewhere in the analysis (i.e.,

they were not included in the input matrix V). A fairer comparison would involve deriving marker genes solely from the GSE datasets that were provided to challenge participants or from published literature.

We performed differential expression analysis using the *voom* normalization [68] and the *limma* package [69]. The top 20 marker genes for each cell type were selected.

For the reference-free scenario, all eight datasets from the DREAM challenge [38] were prefiltered by removing open reading frames, small RNAs, ribosomal, mitochondrial, housekeeping, and cell cycle genes, as well as genes without an assigned symbol. From the remaining set, the 15,000 genes with the highest MAD values were retained for deconvolution.

The number of cell types used for deconvolution was set to the number of cell types present in the ground-truth proportion matrix, plus one additional cell type to account for the unknown cancer component. After random initialization, the optimization procedure was performed using the default Dual Simplex optimization method, yielding matrices W and H . For comparison, the same datasets were deconvolved with Linseed, TOAST, CAM3, and CellDistinguisher following the procedures recommended by their respective manuals. The resulting proportion matrices (H) were compared to the ground truth matrices from the fine-grained DREAM challenge using Pearson correlation and RMSE. Full Pearson correlation matrices for each dataset and method are provided in Additional file 3: Table S3.

For the marker-based scenario, the eight marker expression matrices were deconvolved using our proposed marker-based approach without additional filtering. The number of cell types used for deconvolution was set to the number of cell types present in the ground-truth proportion matrix, plus one additional cell type to account for the unknown cancer component. The resulting proportion matrices (H) were compared to both the published DREAM results and the fine-grained ground truth matrices, again using Pearson correlation and RMSE. Complete Pearson correlation matrices for each dataset and method are presented in Additional file 3: Table S4.

Single-cell RNAseq HNSC dataset clustering

The GSE103322 dataset of head and neck squamous cell carcinoma was extracted from the GEO database [70] and linearized ($V = 2^V$). The top 3000 most variable genes were selected using the Seurat FindVariableFeatures method [44]. For outlier filtering, we applied consequent distance cutoffs, reaching thresholds of *zero_distance* = 0.08, *plane_distance* = 0.2 for cells, and *zero_distance* = 0.048 for genes. This resulted in a dataset of 2647 genes and 5812 cells.

We then performed nonnegative factorization of this dataset via the Dual Simplex approach, with the number of pure components K ranging from 3 to 18. The gradient descent was initialized with a random initialization, with optimization performed according to the default optimization procedure described above. Clusters were then assigned to cells as the row index for the maximum value of the column representing this cell in the matrix H .

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13059-025-03910-1>.

Additional file 1: Supplementary Note 1. The proof of the existence of a first simplex in the sample space of a row-normalized matrix. Supplementary Note 2. The proof of the existence of a complementary simplex in the feature space of a column-normalized matrix. Supplementary Note 3. Definition of Sinkhorn scaling transformation. The relation between two simplexes of the Sinkhorn transformed matrix. Supplementary Note 4. Definition of the projection operation for a N -dimensional sample space to a lower K -dimensional space. Vectors R . The projective formulation of NMF/deconvolution in this space. Supplementary Note 5. Definition of the projection operation for a M -dimensional feature space to a lower K -dimensional space. Vectors S . The projective formulation of NMF/deconvolution in this space. Supplementary Note 6. $(K-1)$ -dimensional structure of a simplex in sample space leads to additional constraint for projection (R) vectors we want to maintain. Supplementary Note 7. $(K-1)$ -dimensional structure of a simplex in feature space leads to additional constraint for projection vectors we want to maintain. Supplementary Note 8. Definition of the same problems for Sinkhorn transformed matrices. The constant relation allows two independent projection optimization problems to join into a single optimization problem. Supplementary Note 9. Sinkhorn transformed matrices inherently satisfy additional constraints we want to define, so singular vectors are a possible solution for projection minimization. The Dual Simplex Theorem. Supplementary Note 10. Gradient descent training formulations to search simplex corners in projected space. Update rules, hyperparameters, and initialization strategies. Geometrical relation between simplexes. Supplementary Note 11. Definition of reverse Sinkhorn transformation to return to the original space solution matrices W and H . Supplementary Note 12. Application notes and guidance on data prefiltering.

Additional file 2: Fig. S1. Projective NMF solution approach. The general schema of projection NMF approach. Fig. S2. Visualization of projection optimization solutions. Visualization of the possible projection vectors representing simplex points. Fig. S3. Additional NMF experiments. Additional NMF experiments with simulated data. Fig. S4. Further analysis of the bulk-HNSC dataset. Visualization of multiple deconvolution solutions and pathway enrichment of extracted marker genes. Fig. S5. Marker-based deconvolution with the DualSimplex approach. Marker-based deconvolution performance on benchmark datasets. Fig. S6. DREAM challenge correlation matrices. Fig. S7. Single-cell data analysis with the DualSimplex approach. Fig. S8. Multiple NMF solutions. Analysis of method behavior in NMF formulations with multiple valid solutions.

Additional file 3: Table S1. Genes filtering for bulk-HNSC dataset. Table S2. Filtering steps applied to the bulk-HNSC dataset. Table S3. DREAM challenge correlation matrices for reference-free scenario. Table S4. DREAM challenge correlation matrices for marker-based scenario.

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Authors' contributions

D.K. wrote the main manuscript text, prepared figures 1-7, prepared supplementary figures 1-8, participated in algorithm development, participated in code development. E.A. participated in algorithm development, participated in code development. A.S. prepared supplementary figures 4 and 7, participated in code development. M.A. supervised the study, wrote the main manuscript text and participated in algorithm development.

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Data availability

The GEO datasets analyzed in this study were downloaded using the following accession codes: GSE19830, GSE11058, GSE181919, GSE199324, GSE103322.

DREAM Challenge expression matrices, ground-truth proportion matrices, and participant predictions were obtained from their corresponding Synapse repositories: syn21821122, syn21820011, and syn22320329, respectively.

TCGA bulk HNSC expression data and sample annotations were accessed through the GDC Portal: <https://portal.gdc.cancer.gov/projects/TCGA-HNSC>.

Computationally mixed brain data and the original single-cell dataset (VL) were obtained from [36] via the corresponding GitHub repository: <https://github.com/Voineagulab/BrainCellularComposition>.

The DualSimplex R package source code is available at: <https://doi.org/10.5281/zenodo.17795706> and is distributed under the MIT license.

Scripts for generating all figures, including simulated data generation, are provided in a dedicated repository: <https://doi.org/10.5281/zenodo.17843610>, also under the MIT license. All input data files required to run the scripts are included in the same repository, with larger files stored separately at: <https://doi.org/10.5281/zenodo.17793325>.

The RcppArmadillo package is available from the CRAN repository: <https://doi.org/10.32614/CRAN.package.RcppArmadillo>.

Declarations

Ethics approval and consent to participate

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Consent for publication

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Competing interests

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